LOGO DETECTION AND BRAND RECOGNITION

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SESSION 2022/2023

FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
JULY 2023

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BY

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SESSION
2022/2023

THE PROJECT REPORT IS
PREPARED FOR

FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
IN PARTIAL FULFILLMENT
FOR

BACHELOR OF COMPUTER SCIENCE
(HONS) DATA SCIENCE

FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY

JULY 2023

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_____

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ACKNOWLEDGEMENT

I want to express my gratitude and deep appreciation to my supervisor, Dr. Noramiza Binti Hasim, for all of her aid and guidance throughout this project. Without her help and expertise, this study would not have been able to get the findings it did. Next, I want to thank my family and friends for supporting me as I worked on this project and for encouraging me to be positive.

ABSTRACT

Brand exposure is particularly important for sponsors and advertisers who want to measure the effectiveness of their advertising campaigns. Logo detection and recognition is an approach to detect and recognise the logos and brands of sponsors and advertisers in badminton sports. This study will be useful to provide valuable insights into brand exposure and help to measure the effectiveness of advertising the brands in a campaign. In this paper, the focus is to study and analyse existing Convolutional Neural Network (CNN) methods for Brand Logo detection as we compared three different deep learning-based methods (i.e. MobileNetV2, InceptionV3, YOLOv8) for logo detection. The models are trained using images, and their performance is evaluated by feeding a highlight badminton video into the trained models. The significance of evaluating the models' performance using multiple metrics, including Mean Average Precision (mAP), Recall, and Precision, is highlighted. The project intends to demonstrate the efficacy of CNN-based methods for logo detection and brand recognition. The findings from this study contribute to the understanding of the models' capabilities and provide insights into their performance in real-world scenarios.

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Chapter 1 : Introduction

1.1 Project Overview

Logo recognition plays a significant role in various domains, including the world of sports. In particular, the relevance of logo recognition in badminton highlight videos is closely tied to the concept of brand recognition. While viewing badminton highlight videos, viewers are regularly exposed to a brand's logo, which helps them develop an impression of the company in their brains. Businesses must comprehend how consumers view their brand, value their products, and recognise their logo. Even without being expressly given the business's name, logo recognition acts as a visual cue that enables customers to recognise and relate to a brand. Businesses can assess the extent of brand identification and awareness among their target audience by regularly featuring their logo in badminton highlight videos.

A company's attempts to increase logo recognition can be evaluated for efficacy through market research and analysis. The ultimate marketing objective is to encourage the target market to connect the brand's emblem with the item, forging a unique identity that distinguishes it from rivals. Creating a strong brand equity is strongly related to developing a strong logo recognition. The presence of a brand's logo in badminton footage during competitions, sponsorships, or player endorsements increases consumer awareness of the company. Viewers' knowledge and sense of connection with the brand are furthered when they notice the emblem prominently displayed on player jerseys, equipment, or event banners. Consumers are more likely to think about the brand when making purchases as a result of this frequent exposure, which helps solidify the brand's existence in their minds.

1.2 Problem Statement

The problem at hand is the lack of sufficient logo detection in badminton highlight videos and its impact on brand recognition. Currently, there is a dearth of strategies and initiatives aimed at effectively showcasing brand logos during badminton matches and related promotional content. This poses a challenge for businesses seeking to establish a strong presence and identity in the minds of viewers. Without prominent logo visibility, it becomes difficult for companies to gauge the level of brand awareness among the target audience and evaluate the effectiveness of their marketing efforts. Consequently, this hinders the ability to create a distinct brand image and gain a competitive edge in the market. Therefore, complete strategies and techniques are required that emphasise logo recognition in badminton highlight videos so that businesses may boost their brand awareness, engage with customers, and eventually increase sales.

1.3 Project Objectives

The objectives of this project are:

- To study and analyse the existing methods for detecting logos and recognising brands.
- To implement selected algorithms for brand logo recognition.
- To evaluate the performance of the algorithms using a selected custom dataset.

1.4 Project Goals

The primary goal of this project is to compare selected algorithms with a transfer learning technique for logo detection. The project aims to evaluate the performance and effectiveness of these algorithms in detecting and recognizing logos in badminton highlight

videos. By conducting a comparative analysis, the project seeks to identify the most suitable approach for logo detection in the context of brand recognition.

Additionally, the project aims to implement a transfer learning technique that takes advantage of pretrained models. By leveraging the knowledge and features learned from these pretrained models, the technique will be customised and tailored to address the unique challenge of detecting logos in badminton highlight videos. The ultimate goal is to develop a functional prototype that effectively integrates the transfer learning technique, enabling accurate and efficient logo detection and recognition.

1.5 Project Scope

The project's deliverable will be a fully operational implementation of the transfer learning technique for logo detection in badminton highlight videos. This implementation will involve adapting and fine-tuning the pretrained models to specialise in logo detection. By utilising the learned features and patterns from these models, the technique will enable the identification and localization of logos within the dynamic and fast-paced environment of badminton highlight videos.

Through this project, a thorough evaluation and comparison of the algorithms, alongside the implementation of the transfer learning technique, will provide valuable insights into the strengths, weaknesses, and performance of each approach. The findings will contribute to the advancement of logo detection techniques and help businesses improve their brand recognition strategies in the domain of badminton video content.

1.6 Project Planning

Phase 1 - TRIMESTER 1 2022/2023														
Activity	**Week**													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Introduction														
Literature Review														
Theoretical Framework														
Research Methodology														
Implementation Plan														
Conclusion														
Prototype Design														
Oral presentation														

Table 1.1 Phase 1 Gantt Chart for trimester 1 2022/2023

Table 1.1 shows the first phase project planning of this research project. Phase 1 of this research project will take place for 14 weeks. From week 1 to week 2, the introduction and the problem statement of this research is explored. To gain a better understanding of the study area, the most recent works in the related domains are reviewed during the next 6 weeks. At the same time in week 6, theoretical frameworks that are going to be implemented for Phase 2 are being focused while also concentrating on research methodology in the next four weeks. The prototype is made continuously on week 9 and 14 following the theoretical framework and research methodology in order to evaluate the effectiveness of the proposed model. Oral presentation on the findings from the FYP1 report writing marks the project's completion on week 13 and 14.

Phase 2 - TRIMESTER 2 2022/2023														
Activity	**Week**													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Introduction														
Literature Review														
Theoretical Framework														
Research Methodology														
Implementations														
Evaluation of Findings														
Conclusion														
Implementation														
Poster Presentation														

Table 1.2 Phase 2 Gantt Chart for trimester 2 2022/2023

Table 1.2 shows the second phase project planning of this research project. The project duration in phase 2 is within 14 weeks same as Phase 1. From week 1 to week 2, the introduction of this research is to be further explored and improved. Over the next 2 to 3 weeks, The most current works in the related fields are again to be further examined to have a deeper understanding of the study area. The following 5 to 6 weeks will concentrate on the theoretical framework as well as the research methodology while implementing the selected models into simulation until week 11. The evaluation of findings will be taken note of on week 12 following the implementations of the model in order to evaluate the effectiveness of the proposed model while writing for the conclusion of this research project. Poster presentation on the research topic marks the project's completion on week 13 and 14.

Chapter 2 : Literature Review

2.1 What is Logo Recognition?

Identifying and separating a particular logo or brand sign from others in an image or video is referred to as logo recognition. To find, locate, and identify logos in diverse visual media, it includes using computer vision methods and algorithms. In order for businesses to create a visual link between their brand and customers, logo recognition is essential to brand awareness and marketing campaigns. Multiple processes are involved in the logo recognition process. To find areas that could contain logos, the picture or video is first examined. This can be done through object detection algorithms that scan the visual content for patterns and shapes resembling logos. Once the potential logo regions are identified, further analysis is performed to verify and classify them as actual logos.

Logo recognition algorithms typically leverage machine learning and deep learning techniques. They are trained on large datasets consisting of annotated images or videos containing various logos. The algorithms develop the ability to recognise and distinguish between many brands by learning to extract distinctive elements and patterns that are unique to certain logos during training.

2.2 Overview

The extracted features of the detected logo regions are compared with the trained patterns from the training dataset throughout the recognition phase. This comparison can be based on a variety of methods, including neural network-based methods, feature extraction, and template matching. The system assesses how closely the detected logo resembles other recognised logos, allowing the brand connected with the logo to be identified.

There are various practical uses for logo recognition. To track brand exposure, evaluate the success of marketing initiatives, and comprehend customer preferences, it is frequently used in advertising and marketing. By locating instances of improper logo use or fake goods, it also aids brand protection initiatives. Additionally, by customising adverts based on recognised logos, logo recognition enables content producers and marketers to offer personalised experiences. The precision and effectiveness of logo recognition systems have been greatly enhanced by developments in computer vision and machine learning. However, obstacles like logo occlusion, size fluctuation, and image quality can still make accurate recognition challenging.

2.3 Related Recent Works

Convolutional neural networks are a subtype of neural networks that are particularly skilled at processing input having a grid-like design, such as pictures. CNNs and ConvNets are other names for convolutional neural networks. For deep learning algorithms, a CNN is a specific kind of network design that is used for tasks like image recognition and pixel data processing. Despite the fact that deep learning uses a variety of neural network types, CNNs remain the preferred network design for recognising and classifying objects. There are several

reasons why CNNs are the preferred network architecture for identifying and recognizing objects. One reason is that CNNs are able to automatically learn a hierarchy of features from raw data, such as pixels in an image. This is done through a process called convolution, where a small, learnable filter is applied to the input data to extract features at different scales and locations. These features are then passed through a series of non-linear transformations, such as activation functions, to extract more complex features. This process is repeated multiple times, forming a hierarchy of features that can be used to classify or recognize objects in the input data.

Another reason why CNNs are effective for object detection and recognition is that they are able to handle large amounts of data and can be trained on large datasets. This is important because object recognition tasks often require a large number of examples in order to learn the variations and variations in object appearance. CNNs are also able to learn from data that has a high level of variability, such as images that are taken from different angles or under different lighting conditions. This makes them well-suited for tasks such as object detection in real-world images.

In recent works of logo detection and recognition, most authors proposed a model or method that is CNN-based. Y. Meng and S. Hou et al (2021) created a strong baseline technique dubbed "obtaining sufficient features for logo" (OSF-Logo) to address the issues with multiscale objects, large-scale categories, viewpoint logos, and part deformation. This method is mainly divided into four parts as shown in the figure below.

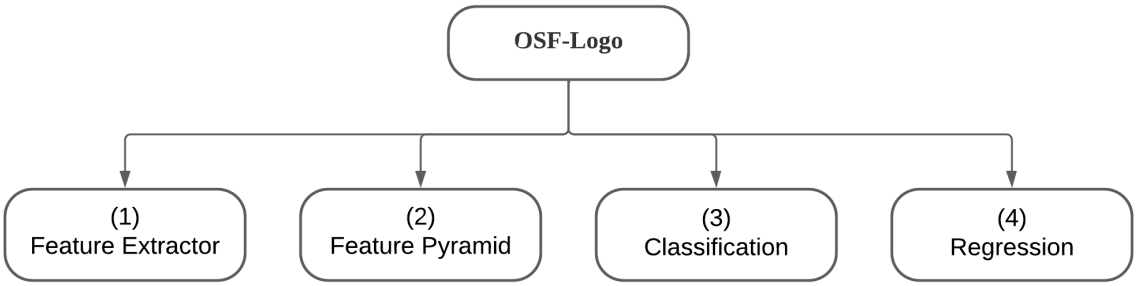


Figure 2.1 four parts of OSF-Logo method. Taken from Y. Meng et. al. (2021)

The first part is Feature Extractor specifically using the Regulated Deformable Convolution (RDC) module with offsets and amplitudes to more convolution layers. Second part is Feature Pyramid, where the authors constructed Content-Aware ReAssembly Features in FPN (FPN-CARAFE) (CFPN) with an up-sampling operator for combining data into a sizable receptive field to detect multi-scale logos. Third part is Classification. With the help of dynamic content perception, the suggested CFPN method achieves feature reorganisation in order to extract the features of small targets and regress large-scale logos. The main idea behind CARAFE is to improve the efficiency and effectiveness of convolutional neural networks (CNNs) for image processing tasks by using a dynamic, content-aware approach to reassembling feature maps. In traditional CNNs, feature maps are typically generated by applying a series of convolutional filters to the input data. These feature maps are then passed through a series of non-linear transformations, such as activation functions, to extract more complex features.

The Kernel Prediction module is used to create the recombined computation's kernel while the Feature Recombination module is to determine the points that correspond to each pixel on the original feature map. Last part is Regression, Y. Meng adds DIoU loss to regression as regression loss for precise logo detection. Similarly, Hang Su and S. Gong et al

(2021) proposed a Multi-Perspective Cross-Class (MPCC) which aligns the feature distribution among synthetic logo images, real logo images, and non-logo object photos by investigating unsupervised domain adaptation. Extensive tests on the publicly available QMUL-OpenLogo benchmark demonstrate the performance advantage of the proposed MPCC method for open logo detection over cutting-edge alternative approaches. The main concept is to apply the understanding of fully supervised logo classes to 1-shot icon supervised logo classes.

Ahmed Alsheikhy et al (2020) proposed an architecture for logo recognition. The authors are interested in effective representations that can be accomplished with a limited set of parameters as they tackle recognition with a small amount of original data. So the authors explored the Densely Connected Convolutional Networks (DenseNet) to get efficient results. A DenseNet is a kind of CNN that uses Dense Blocks to directly connect all layers (with matching feature-map sizes) with one another in order to create dense connections between them.

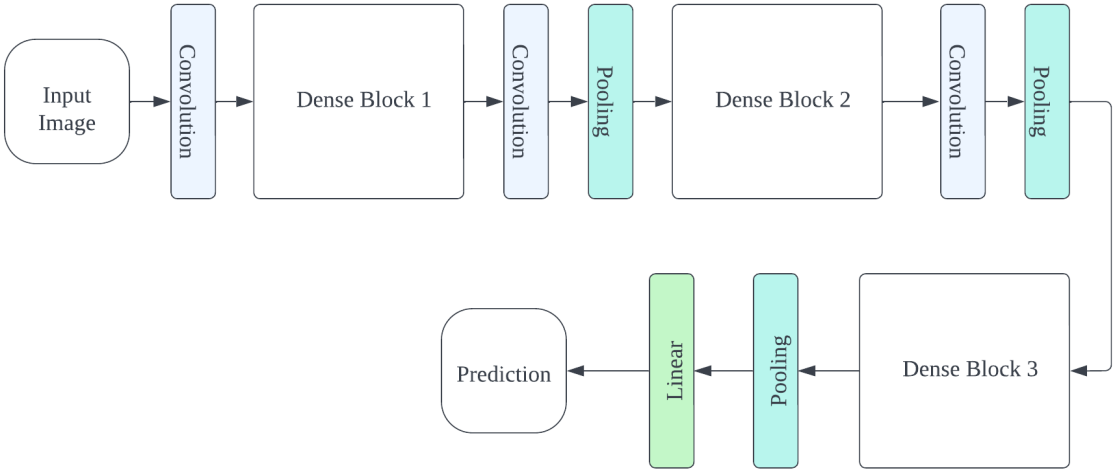


Figure 2.2 DenseNet architecture. Taken from Ahmed et. al. (2020)

A collection of convolutional layers with extensive interconnections is referred to as a dense block. All outputs from earlier layers are given to the current layer as input. Convolution layers are packed into a single dense block, followed by ReLU activations layers and Batch Normalisation Layers which is to reduce the size of feature maps. Instead of using the classes from the original ImageNet dataset, Ahmed et al. used the transfer learning technique to configure the output layer for the logos classes. The reason is to enable the reuse of Deep CNN models that have already been created without the need for extensive computing. Plus it also helps prevent overfitting when accelerating network training. The weights of the custom-made network were initialised with a pre-trained DenseNet model on the ImageNet Dataset. Then, the network was fine-tuned on the FlickerLogo32 training set.

K. Kumar J. P. and Supriya M. C. (2021) also proposed a CNN-based system for logo detection in television programs.The suggested system comes with a pipeline for logo detection and classification in TV broadcast and broadband content that demonstrates how to use the object detection algorithm Single Shot Detector (SSD) on a TV broadcast channel logo dataset that has undergone significant changes to better represent various logo conditions. Generally speaking, the suggested system consists of two pieces. The first section focuses on the extraction of frames, and to create fake data, different data augmentation techniques are used with the retrieved broadcast logo. This data augmentation is done to increase the trained network's coverage of various logo types in terms of size, form, and visibility. In the second part, Single shot detector (SSD) architecture is used for single forward pass detection and classification. SSD employs the VGG16 design without taking into account the fully connected layers. The VGG16 architecture was selected as the base due to its consistently good performance for classification tasks. The number 16 in the name VGG indicates the fact that it is a 16 layers deep neural network (VGGnet). This indicates that the

VGG16 network is quite large, with a total of over 138 million parameters. But instead of using VGG16 fully connected layers, a group of convolutional layers has been added, enabling input size reduction at each layer as well as feature extraction at various scales. The frame that was intercepted is sent to the VGG16 network model, which produces the feature extraction. These features are then sent to several convolutional layers and convolutional filters, which are used to identify and categorise the item.

Amy J.C. Trappey and Charles V. Trappey et al (2022), integrates two deep learning models to achieve the goal of assuring copyrighted Trademark (TM) logo protection. The first model is Logo Detection and Localization (LDL), which is used to clip trademark-like images from intricate online product photography, which may include shots of various product configurations or even wearable clothing exhibited on a human model. For comparison to a database of registered trademarks, only the cropped image of the logo is required. The second model, utilising the cropped logos for the TM similarity measurement (TMSM) model, compares the published TM pictures. The universal logo locator for intelligent logo image cropping is Yolo v4. For intelligent trademark similarity analysis, the triplet convolutional neural network ResNet50 model is used. The models are trained using an image dataset that combines LogoDet-3k picture samples with photos retrieved from the internet.

2.3.1 Summary of Related works and the results

Table 2.1 Related works and Results

No	Author	Method Used	Remark	Result
1.	Y. Meng and S. Hou et al (2021)	Propose OSF-Logo Regulated Deformable Convolution (RDC) ResNet50 and Content-Aware ReAssembly Features (CARAFE) in FPN (CFPN) as backbone network	The authors conduct extensive tests and assessments on five significant benchmark datasets, including the FlickrLogos-32, Logo-Det-3 k-100, QMUL-OpenLogo, Necessities, and Logo-Det-3 k-1000 datasets, to confirm the effectiveness of the technique.	Logo-Det-3k-1000: 89.9% mAP. FlickrLogos-32 : 87% mAP Necessities : 84% Logo-Det-3 k-100 : 80.9% mAP QMUL-OpenLogo: 53.3% mAP
2.	Hang Su and S. Gong et al (2021)	Propose a Multi-Perspective Cross-Class (MPCC) domain adaptation method	For model evaluation, two powerful object detection models, YOLOv2 and Faster R-CNN, along with two synthetic data generation techniques, SCL and CAL, were	1-shot supervised: SCL in MPCC 23.4% mAP CAL in MPCC 24.53% mAP

		ResNet-101 based Faster R-CNN as the base for detection model	compared with the MPCC by the authors.	Fully supervised: SCL in MPCC 48.6% CAL in MPCC 49.41% mAP
3.	Ahmed Alsheikhy et al (2020)	Propose a DenseNet architecture for logo recognition	The FlickrLogos-32 dataset for logo recognition was utilised for training and testing to assess the proposed model. The recommended model achieves high classification accuracy and cutting-edge performance.	92.8 % of average accuracy was achieved
4.	K. Kumar J. P. and Supriya M. C. (2021)	Propose a CNN-based system for logo detection in television programs Data Augmentation methods	The result of each genre of the proposed SSD model is compared with the YOLO method. The results indicate that single shot detection (SSD) is successful in tackling the issue of content piracy in real-world datasets.	Proposed System: Precision is range between 80.5% and 86.7% Accuracy is range between 81% and 85% YOLO method: Precision is range between 74.5% and

		SSD-VGG16		84.5% Accuracy is range between 72.5% and 82%
5.	Amy J.C. Trappey and Charles V. Trappey et al (2022)	Propose two deep learning models: Logo Detection and Localization (LDL) and TM similarity measurement (TMSM) model YOLOv4 as LDL ResNet50 as backbone of TMSM	The system performance is statistically evaluated using the FlickrLogos-32 dataset and consumer product photos that were taken from an e-commerce portal. The proposed system achieves high recall and precision when matching top 10 TM classes.	Recall : 46% Precision : 91%
6.	Andrey Kuznetsov and Andrey V.	Proposed two-stage approach, using MobileNet and YOLO v3	The research conducts an experimental study using different combinations of backbone and detector CNNs. The goal is to evaluate the performance of	The reported evaluation metrics include mean Average Precision (mAP) of 0.74 and

	Savchenko (2020)		various models on the SportLogo dataset. The paper specifically mentions the MobileNet architecture combined with YOLO v3 as the best-performing solution on the dataset.	Recall of 0.87. These results indicate the effectiveness of the proposed approach in detecting logos in sports scenes, considering the challenges presented by occlusions, view changes, and style variations.
7.	Salma Sahel et al (2021)	Utilises pre-trained object detection models to enhance the performance of logo detection tasks (RCNN and Faster RCNN)	To apply emerging deep learning techniques to perform brand and logo recognition tasks. The study uses the FlickrLogos-32 dataset, which is a common public dataset for logo detection and brand recognition from real-world product images. This dataset serves as the basis for evaluating the models.	Precision: 0.948 Recall: 0.829 Accuracy(RCNN): 0.998 mAP(RCNN): 0.65 Accuracy (FR-CNN): 0.936 mAP(FR-CNN): 0.74

2.4 Challenges and Research Gaps

Amy J.C. Trappey et al (2022) faced the biggest obstacle, which was locating and categorising the logos in the photographs of product displays that are provided. The input photos are cluttered with noise and lack background and logo objects. According to the authors, it may be difficult to locate and categorise logos that are tilted, reversed, or partially obscured. The authors highlighted that by utilising more sophisticated models or enhancing the training picture sample process, future studies might focus on narrowing the similarity gap between trademark templates and recognised logos. To cover a larger range of logo image types across different websites, the described technique may also be utilised for partially automated picture data labelling. By utilising the proposed image collector module to gather pictures from various sources, these photographs, which either contain existing logo types or non-existent (new) logo types, may be rapidly and accurately recognised and tagged as a growing and adaptable logo training set. Consequently, the future work may be rapidly enlarged by utilising the recommended approach in order to create an intelligent trademark patrol and protection system for several enormous e-commerce platforms.

Contrarily, according to Y. Meng et al. (2021), following upsampling, the feature maps with strong semantics from the higher layers are integrated with the lower layers, making it feasible to find tiny targets.. Nearest neighbour and bilinear interpolations are the two most often utilised feature up-sampling operators. However, because they both only consider the sub-pixel region, none of these two methods can effectively capture the rich semantic information required for logo identification tasks. Authors also mention that for increasingly intricate and rich logo cases, authors will concentrate on creating lighter and more efficient logo detection networks.

The project goals highlight several research gaps in the field of logo detection especially in badminton highlight videos. First and foremost, a thorough comparison of logo detection algorithms created especially for badminton highlight videos is required. The effectiveness, accuracy, and efficiency of existing algorithms must be assessed in relation to problems such as logo occlusion, size change, and logo movement. The best algorithms for logo detection and recognition in the dynamic environment of badminton highlight videos will be found with the aid of this investigation. The application of transfer learning algorithms for logo detection and recognition in badminton highlight video is another area where research is lacking. The best methods, pretrained models, and fine-tuning tactics that can improve logo detection and recognition performance need to be identified through further research. Considering the advantages and drawbacks of transfer learning in this situation will aid in the creation of more effective logo detection models.

Another area of research needed is the production and annotation of appropriate datasets. It is necessary to create varied and thorough datasets that include various brand logo sponsors, logo changes, and real-world situations. Furthermore, it is essential to create trustworthy techniques for precisely tagging logos in these custom dataset. Taking care of these issues will make it easier to train and test logo identification algorithms in badminton highlight videos.

Chapter 3 : Theoretical Framework

3.1 General framework

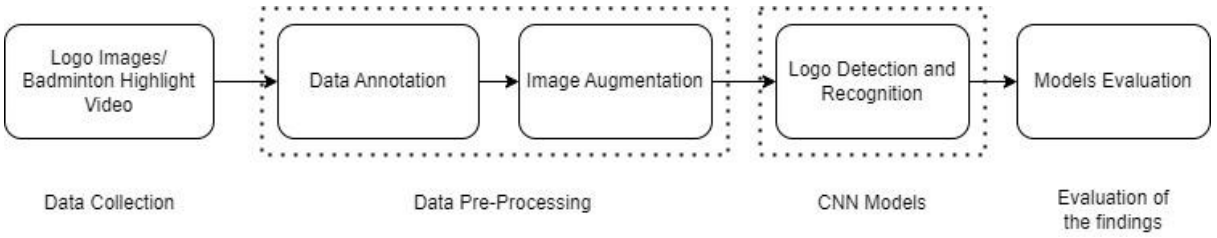


Figure 3.1 General Framework

Our methodology involves training a deep learning model using images to recognize specific patterns and objects. The dataset consists of thousands of logo images that are carefully curated and annotated to provide the necessary information for the model to learn from. The existing convolutional neural network (CNN) models that are being used are well-suited for the recognition tasks due to its ability to extract features and patterns from images. Once the model is trained, the model will be implemented on a highlight badminton video to showcase its performance. The video will contain several scenes that feature the logos the model was trained on, and the model will be able to identify and highlight those logos in real-time. This will demonstrate the effectiveness and accuracy of the model in a practical and visually engaging way.

3.2 Data Preprocessing

There are a lot of general logo dataset like Web logo 2M and Flickr logos, but due to the absence of local badminton sponsor's logos in the general logo datasets, a custom dataset is created and the images of the custom dataset is collected from Google Images. The sponsor

logos for the badminton competition, which Malaysia is hosting, are being added to this dataset, along with labels for the accompanying photos. The second phase of our dataset generation was adding a picture label or annotation. In order to complete this procedure, a labelling tool named "labelImg" is utilised, which is launched as a Python script from the location containing the input picture directory and a text file containing the names of all the necessary classes and stored in YOLO format. All of the picture classes may be easily labelled using this tool.

3.3 Logo Detection and Recognition model

Models
InceptionV3
MobileNetV2
YOLOv8

Table 3.1 Methods

The approaches or models that will be used in this project are as shown in Table 3.1, and the assessment will be conducted by contrasting each of the findings.

3.3.1 InceptionV3

Convolutional neural network model InceptionV3 was created by Google in 2015. It is a design for a deep neural network that is mostly employed for image recognition and classification applications. The 48-layer InceptionV3 model is an adaptation of the original Inception architecture, InceptionV1 which was introduced as GoogLeNet in 2014. On numerous benchmark datasets, including ImageNet, it is renowned for achieving state-of-the-art performance.

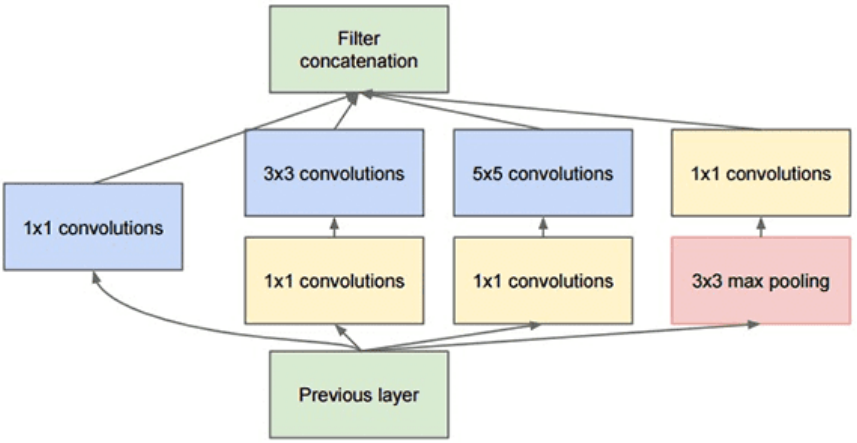


Figure 3.2 The original Inception module used in GoogLeNet. Taken from Szegedy (2014)

At its core, InceptionV3 is made up of several convolutional layers that take input pictures and extract characteristics from them. The addition of Inception modules, which include parallel branches of convolutional filters with various sizes (1x1, 3x3, 5x5) and pooling processes, enhances these convolutional layers. Due to its distinctive architecture, the network is better able to recognise objects of varying sizes inside pictures by capturing characteristics at multiple scales. Auxiliary classifiers are included in InceptionV3's network to alleviate the vanishing gradient issue and encourage better discriminative feature learning. Fully connected layers and a softmax layer for classification round out the structure. Pretrained weights from training on large-scale image datasets are often used to initialise InceptionV3, enabling the network to swiftly adapt to new categorization tasks. InceptionV3 has excelled in image classification, object identification, and image segmentation tasks because of its delicate balancing of complexity and computational effectiveness.

3.3.2 MobileNetV2

MobileNetV2 is a popular convolutional neural network architecture that was introduced in 2018. A state-of-the-art deep convolutional neural network architecture created

exclusively for mobile and embedded vision applications is its framework. It seeks to attain high accuracy while utilising the fewest number of computer resources and models possible.

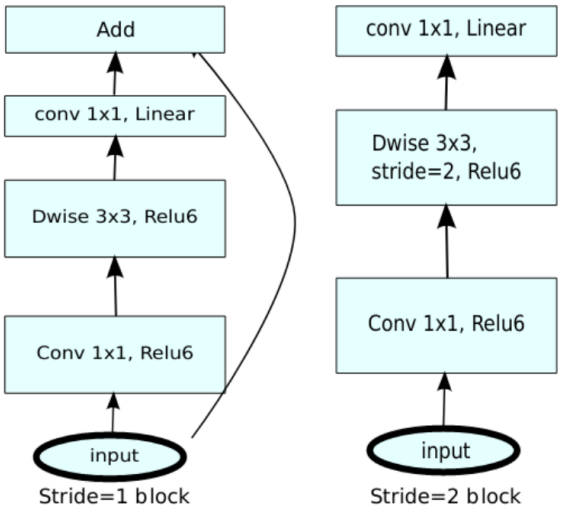


Figure 3.3 MobileNetV2. Taken from Sandler (2018)

Starting with an initial set of convolutional layers that extracts fundamental information from input images, MobileNetV2 is made up of many important parts. The inverted residual block, which consists of a bottleneck layer and depth-wise separable convolutions, is the main construction component. These building components allow for effective computing and capture correlations in both space and across channels. Squeeze-and-excitation (SE) blocks improve feature attention and recalibration while expansion layers expand the representation capacity. A softmax layer for probability distribution over classes is the network's final fully connected layer after layer for feature aggregation and classification. Because it strikes a compromise between model complexity, accuracy, and computational economy, MobileNetV2 succeeds in contexts with limited resources. It provides a strong foundation for a number of computer vision applications, such as picture classification, object recognition, and semantic segmentation in mobile and embedded devices.

3.3.3 *YOLOv8*

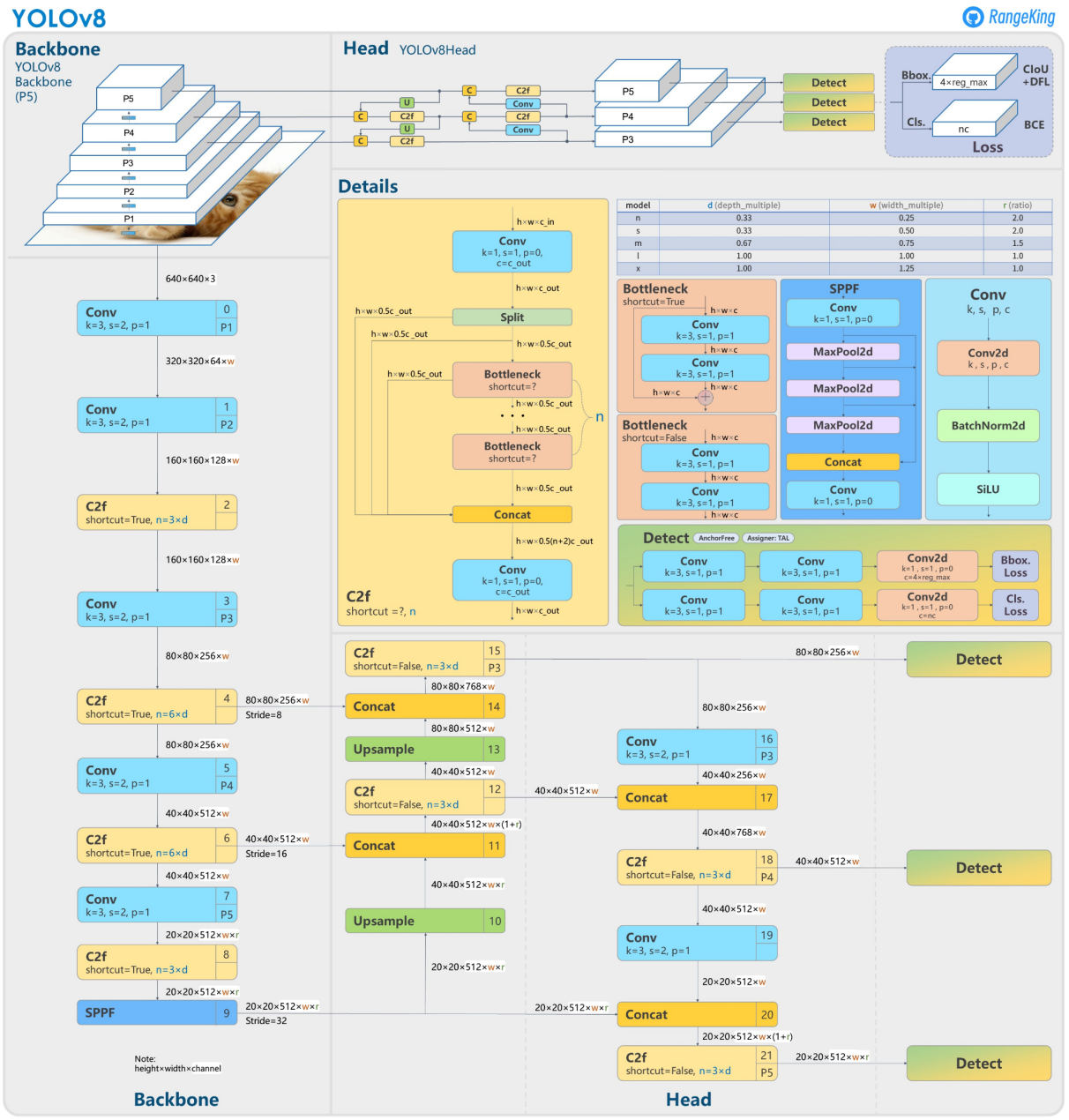


Figure 3.4 YOLOv8 model structure. Visualisation made by GitHub user RangeKing

Glenn Jocher, the CEO and creator of Ultralytics, created YOLOv8, the most recent version of the YOLO (You Only Look Once) object detection paradigm. A well-known business that specialises in creating and implementing deep learning models for computer

vision jobs is Ultralytics. In comparison to its predecessors, YOLOv8 marked a considerable improvement in accuracy, speed, and efficiency when it was shown to the public in March 2022. YOLOv8 has been known for its ability to quickly and effectively identify items across a wide variety of classes thanks to its real-time object detection capabilities. With the launch of YOLOv8, an important step has been made in the development of object identification models, demonstrating the continued developments in computer vision and deep learning.

The primary component of YOLOv8 is a single neural network that predicts bounding boxes and class probabilities for several objects in an input image concurrently. The framework consists of a number of crucial elements, beginning with input processing, which divides the picture into a grid of cells in charge of identifying items in their appropriate areas. A backbone network, like Darknet or CSPDarknet, uses convolutional layers, residual blocks, and downsampling layers to capture both low-level and high-level information as it extracts hierarchical features from the input picture. The neck design, like PANet or FPN, combines characteristics from many scales and levels to increase the network's capacity for handling objects of various sizes and boost detection precision.

Convolutional layers, objectness prediction layers, and class prediction layers are employed by the detection head to analyse the fused data and produce predictions for object presence and assign class probabilities. Scales and anchors improve the precision of object localization and detection. Then, Non-Maximum Suppression (NMS) is used to remove unnecessary bounding boxes and choose the most certain and precise detections. Rescaling the bounding box coordinates and applying confidence criteria are the final steps in post-processing. With its ability to balance speed and accuracy in real-time object recognition tasks, YOLOv8 is ideally suited for applications that need fast and precise real-time object detection in both still photos and moving videos.

3.4 Evaluation Metrics

3.4.1 Precision

Precision is a measure of the accuracy of a logo detection and recognition system. It is determined by dividing the total number of genuine positive detections by the number of positive detections the system produced. False positive detections are when the system wrongly identifies or categorises a logo or brand, whereas true positive detections are when the system correctly detects and categorises a logo or brand.

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{Relevant\ Images}{Total\ Images\ Retrieved}$$

Precision is often used as a performance metric for logo detection and recognition systems because it can give a good indication of the accuracy of the system. While a system with low accuracy may have a high number of false positives, a system with high precision can accurately categorise a significant portion of the logos it detects. Using accuracy as a performance metric has a number of benefits. One benefit is that it is simple to compare and comprehend across many platforms. Its sensitivity to false positives is another benefit, which might be crucial in situations where the cost of false positives is considerable.

3.4.2 Recall

Recall is a measure of the completeness of a logo detection and recognition system. It is calculated by dividing the total number of authentic logos contained in the data by the number of authentic positive detections. False negative detections, on the other hand, occur when the system fails to recognise or categorise a logo or brand that is present in the picture. True positive detections, on the other hand, relate to situations where the system successfully detects and categorises a logo or brand.

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{Relevant\ Images}{Total\ Relevant\ Images\ in\ Data}$$

Recall is often used as a performance metric for logo detection and recognition systems because it can give a good indication of the completeness of the system. A system with high recall is able to detect a large proportion of the logos present in the data, while a system with low recall may have a high rate of false negatives. There are several advantages to using recall as a performance metric. One advantage is that it is sensitive to false negatives, which can be particularly important in applications where missing a logo or brand can have significant consequences. Another advantage is that it is easy to interpret and compare across different systems.

3.4.3 Mean Average Precision (mAP)

Mean Average Precision (mAP) is a measure of the accuracy of a logo detection and recognition system. It is a commonly used metric in object detection tasks and represents the mean of the system's precision ratings at various recall thresholds. After ranking the predictions by descending confidence ratings and comparing the predicted bounding boxes to the actual bounding boxes, it is calculated by averaging the precision at each recall level.

$$\text{mAP} = \frac{1}{n} \sum_{k=1}^{k=n} AP_k$$

Where:

AP_k = the AP of classes k

n = the number of classes

mAP, which is sensitive to both true positives and false positives, is a measurement of how well the system can identify and categorise logos in a picture. A high mAP score means that a significant fraction of the logos in the image can be accurately identified and categorised by the system. One of the key benefits of utilising mAP as a performance metric is that it considers both precision and recall, and it does so across various recall levels, providing a deeper insight of the system performance. It is also commonly used in object detection tasks, making it simple to assess how well various systems perform against one another.

Chapter 4 : Research Methodology

4.1 Components and Software

Table 4.1 shows the processing unit that will be used in this project for both implementing model training and testing and also will be used for report writing and documentation.

Table 4.1 Processing unit Specifications

Brand	**Model**	**Specification**
Asus	ROG Strix G G531	Windows 11, 9th Gen Intel Core i7-9750H With GeForce RTX 2060, 16GB RAM, 512GB Storage

Software's like PyCharm Professional in Jetbrains and Anaconda will be utilised and operated to develop our framework. Anaconda manages the packages and virtual environments, whereas PyCharm are Integrated Development Environments (IDEs) for building a system. Anaconda will be used to create a virtual environment with a specific dependency to run the algorithm. PyCharm is used to view input and output. All software mentioned are free to use but for the PyCharm Professional, it needs to use the University's email to get the JetBrains Student Pack License for free until graduation.

4.1.1 InceptionV3 and MobileNetV2 Setup

For the setup, multiple frameworks and tools were used to implement InceptionV3 and MobileNetV2. The models were created and trained using TensorFlow and Keras, two well-known deep learning frameworks. For building the network architecture, loading pretrained models, and fine-tuning them for the particular object recognition or classification tasks, it offered a user-friendly and effective interface. We utilised Sklearn, a well-known machine learning toolkit, to assess the models' performance. The models' accuracy, precision, recall, and F1-score were evaluated using a variety of metrics, including the classification report and confusion matrix, which were calculated using Sklearn's metrics module. The usage of pandas, enabled for efficient data handling and modification, which improved the data analysis and visualisation components. Popular data visualisation packages Seaborn and Matplotlib made it possible to create instructive plots, charts, and visualisations to convey the results of the training such as Loss function and accuracy of Train and validation in an approachable manner. Last but not least, efficient numerical operations and array manipulation made use of numpy, a core package for numerical computing, made data processing and model training possible. Together, these frameworks and resources provide a thorough and effective environment for putting InceptionV3 and MobileNetV2 into use and assessing them, allowing for reliable analysis and interpretation of the models' performance.

4.1.2 YOLOv8 Setup

The installation of the Ultralytics package at the Anaconda prompt is essential since it will install all the libraries needed to run YOLOv8 after changing the directory to the YOLOv8 environment. By default, this will install the PyTorch CPU version after installing

the Ultralytics. The GPU version must be installed since using the CPU for training would be excruciatingly slow.

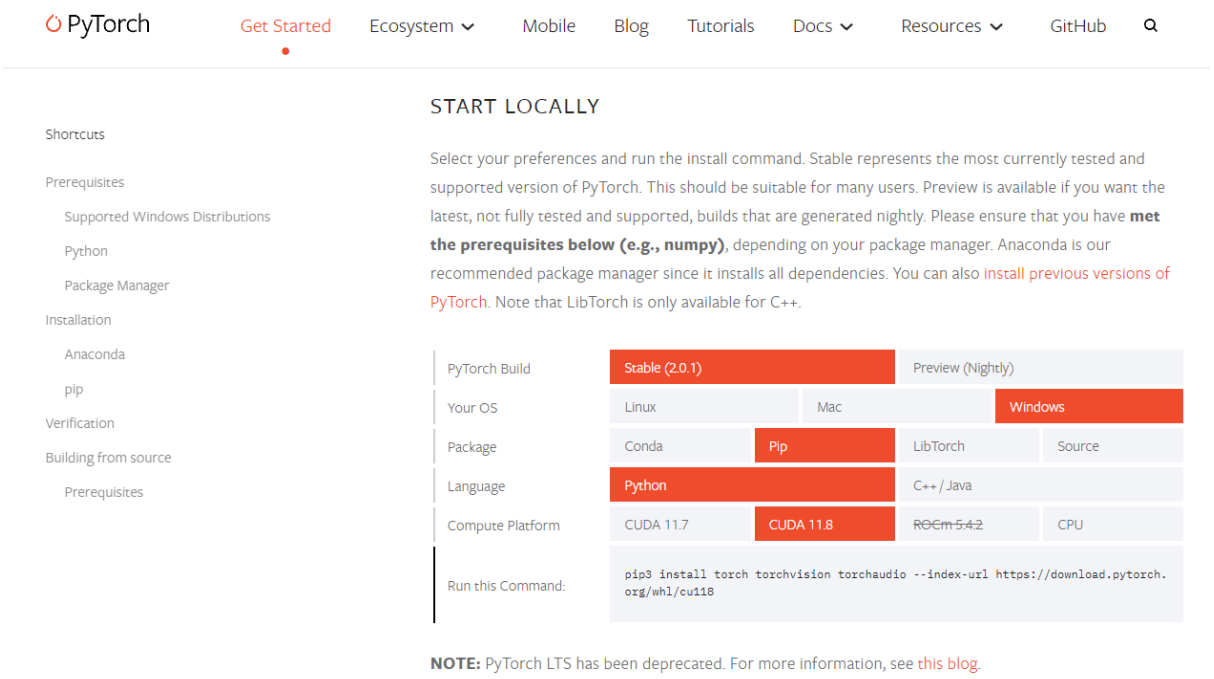


Figure 4.1 PyTorch GPU version installation website

To install the GPU version, you need to visit the official PyTorch website and click on "Get Started" then select your device specs and your preferences then copy the "Run this Command" on the website. Add the '--upgrade' as shown in the figure 4.2 below and run the command in the terminal to install the GPU version of PyTorch.

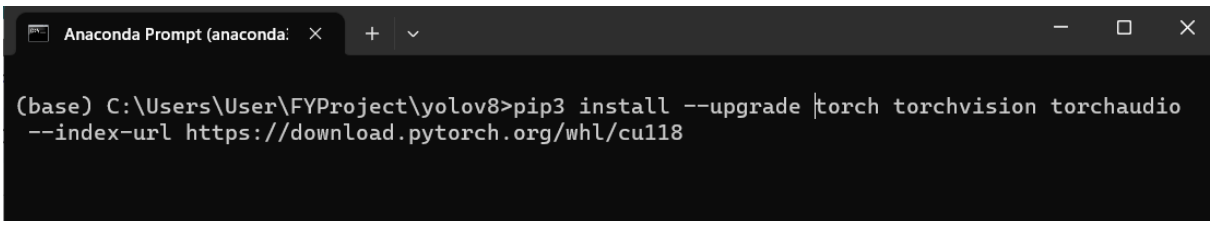


Figure 4.2 PyTorch GPU version installation command

4.2 Dataset

The dataset used in this study was originally sourced from Kaggle, contributed by a user named Hamza. The dataset is then revamped by selecting logos specifically well-known brands in Malaysia. Additionally, logos of brands that are prominent sponsors of BWF badminton events are included. To curate the dataset, the images are collected from Google Images, focusing on 40 classes of logos representing brands that are widely recognized in Malaysia. The dataset is customised to include logos relevant to the Malaysian context and incorporating brands associated with BWF badminton events, in order to enhance the dataset's applicability and relevance to the local domain. The logo for data augmentation, which serves as a regularizer and helps to lessen overfitting while training the suggested model, is represented by 100 distinct graphics for each of the classes. The dataset is made up of 4000 photos in total.

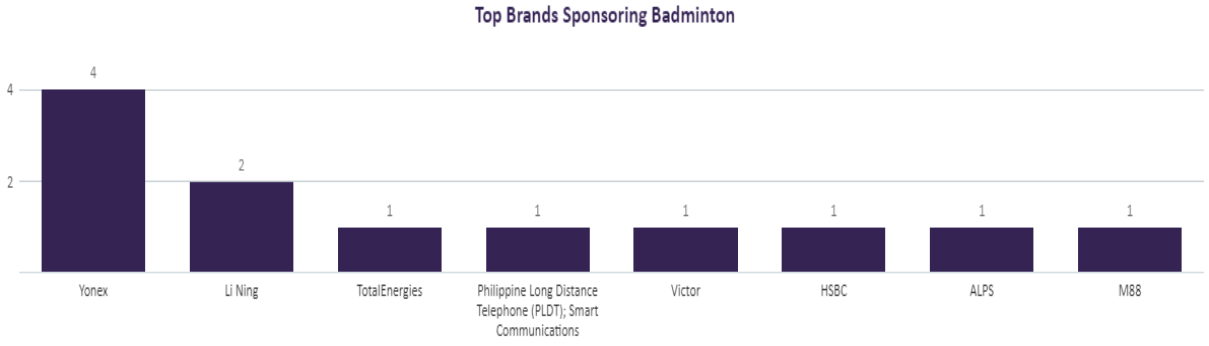


Figure 4.3 Top Brands Sponsoring Badminton. Taken from GlobalData

The four companies selected to be included in the custom dataset are Yonex, TotalEnergies, Victor, and HSBC since they are the leading sponsors of badminton tournaments globally, particularly in Malaysia, as shown in Figure 4.1. For Li-Ning, the majority of sponsorships occur when China is the event's host nation, however they may be added to the custom dataset if a bigger dataset is required. Petronas is also the principal

sponsor with Malaysia serving as the host nation. Some of The BWF World Championships are sponsored by Toyota too. Numerous more companies, particularly in Malaysia, support the badminton competition, including LG and Astro. When building the broader dataset, a bespoke dataset will include these two brands. The sponsorship of events held outside of Malaysia, such as the Indonesia Open and the England Open, will also be taken into consideration for inclusion in the dataset.

4.3 Data Preprocessing

4.3.1 Data Annotation

In general, the term "data annotation" refers to the labelling of data. Although their applications might vary depending on the sector or use case, data annotation and data labelling are frequently used interchangeably. Labelled data emphasises the features, characteristics, or categories of the data that may be examined for trends that aid in target prediction. Bounding boxes, semantic segmentation, landmarks, polygons, cuboids, and polylines are just a few examples of the various sorts of annotation approaches. For this project, bounding boxes used for the process of image annotation as the tool used for labelling is "labelImg" created by Tzutalin in 2015.

First and foremost, download the image annotation tool from GitHub and then install dependencies as shown in the figure 4.2 in Anaconda.

```
conda install pyqt=5
conda install -c anaconda lxml
pyrcc5 -o libs/resources.py resources.qrc
python labelImg.py
python labelImg.py [IMAGE_PATH] [PRE-DEFINED CLASS FILE]
```

Figure 4.4 Dependencies

The virtual environment made for this "labelimg" is activated in order to install the dependencies. Then install the first line of the dependencies, such as the command shown below.

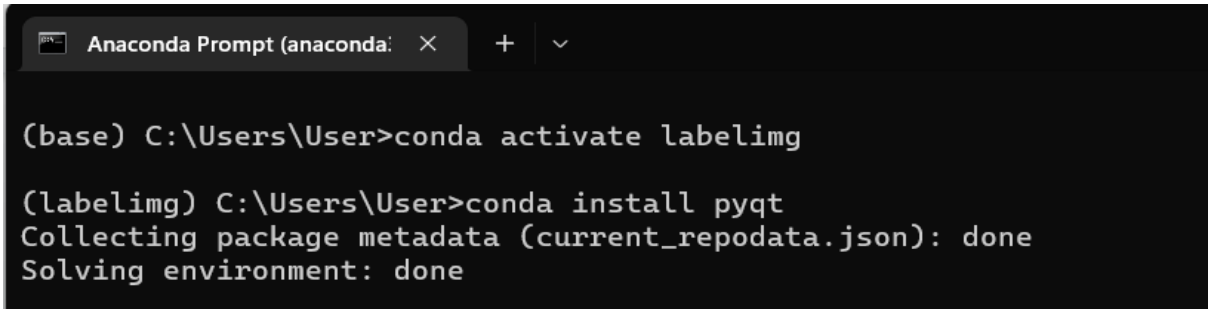


Figure 4.5 Command prompt 1

The names of the classes are then prepared, along with class.txt. Without a gap between words, one in each line of a class, for example; TotalEnergies or Total_Energies. As this project is working with 40 classes, the 40 names of the classes need to be written down as shown in the text file below.

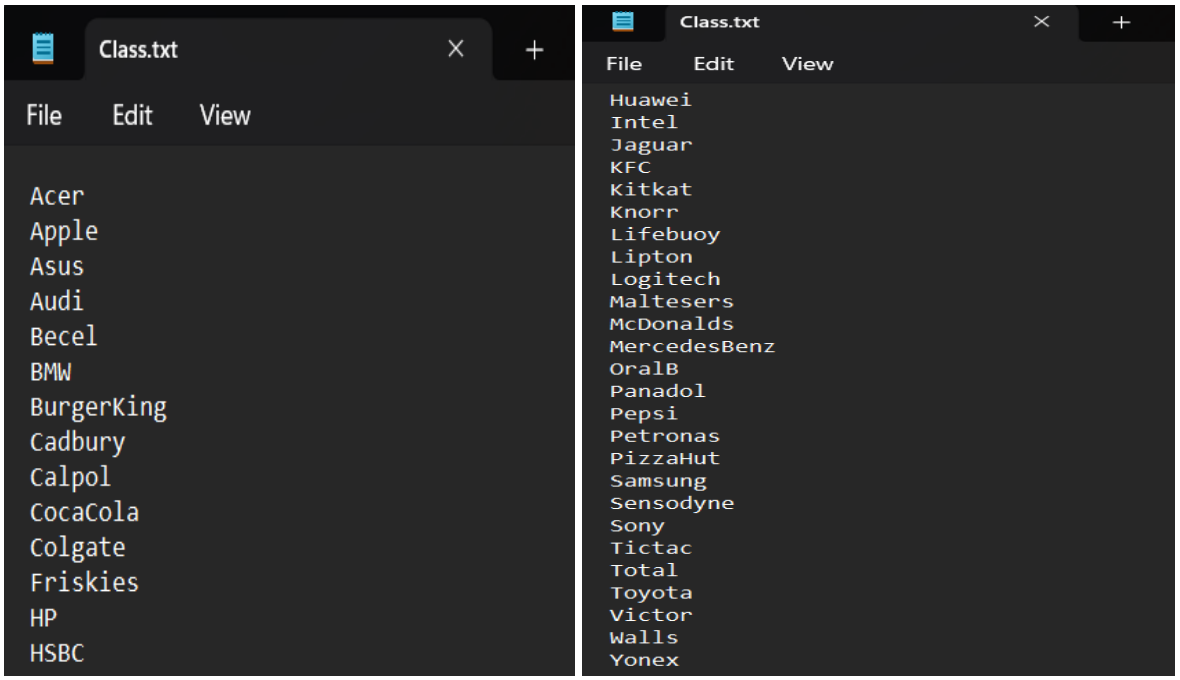


Figure 4.6 Classes text file

After that, go back to Anaconda Prompt and navigate to the path directory of the "labelimg" tool. Then install the second line of dependencies and lxml. The last line in the dependencies is the python script to run the labelImg tool. To run it, replace [IMAGE_PATH] with the image path of the custom dataset and [PRE_DEFINED CLASS FILE] with the path of the class text file class.txt as the command shown below.

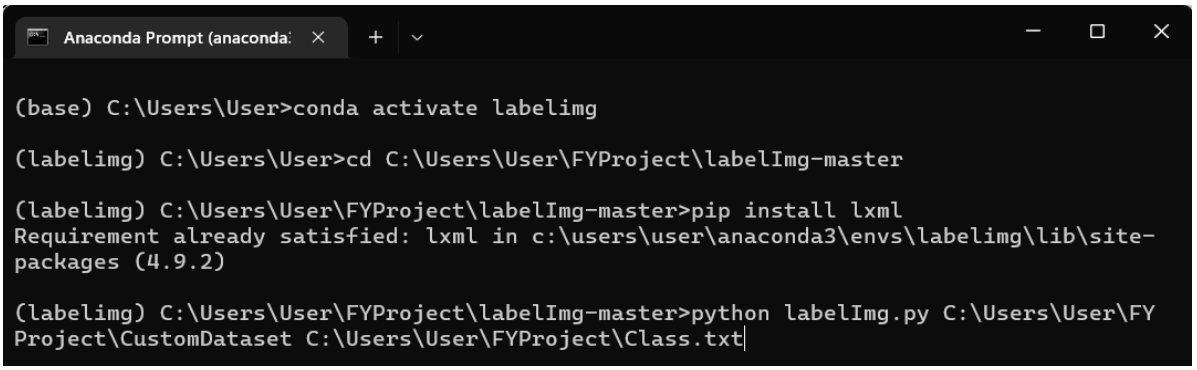


Figure 4.7 Command prompt 2

Following execution of the labelImg.py Python script from the location containing the input photos folder and the classes file, as shown below, the ‘labelImg’ tool GUI is popped out. Next, design the logo's bounding box to name each picture.



Figure 4.8 LabelImg tool

Now that the images have been labelled, the label data is saved in the YOLO format, as the model that will be used is YOLO. The outputs for them look like as shown below.

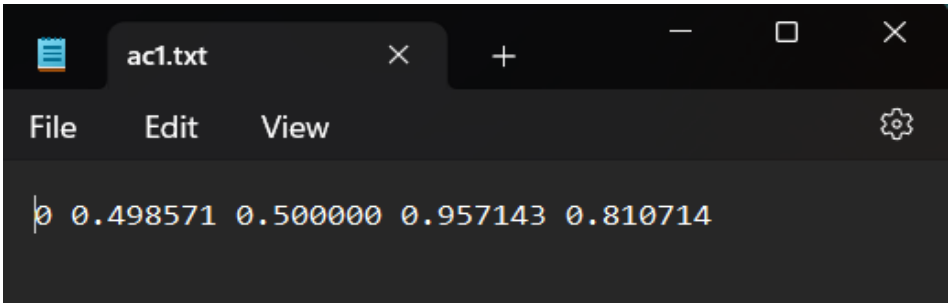


Figure 4.9 Class label text file

As shown in figure 4.7, there are 5 values in the YOLO label file. The first number is the class index id, which spans from 0 (zero) to Classes-1, as stated in the class.txt file. The index begins at zero. Therefore, the value 0 here indicates that it is the first class, or "Acer." The bounding box's height and breadth, along with the object's coordinates, make up the file's last four values. The remaining 4 parameters' values will be normalised in the label file such that they range from 0 to 1. One item's label resembles the one in the image below.

<object-class> <x_center> <y_center> <width> <height>

Figure 4.10 YOLO Label of object

4.3.2 Split dataset

After annotation of the dataset with the location and class of the logos. A ratio of 90% to 10% is used to divide the dataset into training and validation sets, to ensure that the model is trained and evaluated on a representative sample of the data.

The training set is used to train and fine-tune the proposed models. The test set is reserved for evaluating the performance of the final model. This approach is commonly used to ensure that the model is tested on unseen data and that the results are a reliable indication of its performance on new data. The dataset is randomly divided into training and test sets in order to guarantee that they are representative of the complete dataset. The randomization process is done to ensure that the dataset is not biassed towards a particular type of logo. This approach is used to make sure that the model generalises well to new data, and not just memorise the training data. By using this approach, the model is trained and evaluated on a representative sample of the data and it enables estimation of the proposed model's performance on fresh, unused data. The efficiency of the suggested methodology in finding and identifying logos in badminton sports videos is determined using the evaluation's findings.

4.4 Logo Detection and Recognition model

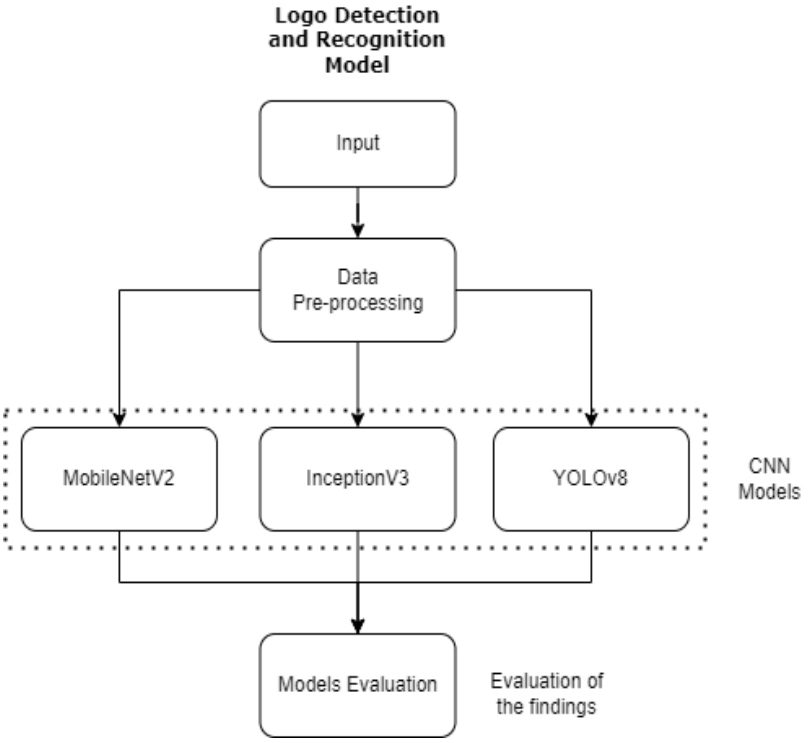


Figure 4.11 Logo Detection and Recognition model Framework

A methodical approach is used in the process for implementing InceptionV3 and MobileNetV2. The initial stage is to precisely specify the study challenge, objectives, and desired performance measures, as well as whether these models are for the object detection or classification task. The chosen custom dataset takes into account its applicability and the presence of labelled or annotated pictures. To increase variety, the dataset is preprocessed using techniques including scaling, normalisation, and data augmentation. By defining the amount of layers, feature maps, and adjustments to meet the project's goals, InceptionV3 and MobileNetV2 are set up. InceptionV3 and MobileNetV2 model weights are loaded and modified for the project-specific objective to use transfer learning. The training process is described, including the regularisation methods, learning rate, batch size, and optimizer. In

the assessment phase, the trained InceptionV3 and MobileNetV2 models are tested on validation and test datasets, performance metrics are reported, and results are compared against benchmarks. Results are evaluated, and InceptionV3 and MobileNetV2's advantages and disadvantages are presented.

For YOLOv8, it entails using a methodical strategy to handle the current object detection task. In order to evaluate the model's efficacy, performance measures, the unique object identification task, and dataset are all specified in the research problem's clearly stated first step. The gathered dataset, labelled bounding box annotations, is typical of the target object types. To increase the dataset's variety and resilience, preprocessing techniques including scaling, normalisation, and augmentation are used. The number of layers, feature maps, anchor boxes, and other hyperparameters are taken into account while configuring the YOLOv8 model to meet the needs of the research challenge. The prepared dataset is then used to train the model, which is subsequently optimised using an appropriate loss function. To get the best performance, a variety of optimisation techniques, learning rates, and batch sizes are tested. On validation and test datasets, the trained YOLOv8 model is assessed using preset metrics, and its outcomes are contrasted with those of other models. The acquired findings are evaluated and discussed, highlighting YOLOv8's advantages, disadvantages, and potential areas for development in the particular object detection task.

Chapter 5 : Implementation

5.1 Data Pre-processing

Rescaling, data augmentation, and normalisation were phases in the data pre-processing procedure for both InceptionV3 and MobileNetV2. The pre-processing techniques from the “tensorflow.keras” “ImageDataGenerator” module was used. The photos were first scaled between 0 and 1 by dividing their pixel values by 255 using the “rescale” option. In order to optimise model convergence, this phase makes sure that the pixel values are contained within a defined range.

Techniques for enhancing the variety of the training data and enhancing the generalizability of the model were also used. Additional training examples with inverted copies of the pictures were produced using horizontal and vertical flipping (horizontal_flip=True and vertical_flip=True). The training data was made more variable using shear transformations (shear_range=0.2) and random zooming (zoom_range=0.2). The dataset was expanded by applying random rotations within a predetermined degree range (rotation_range=40).

Additionally, feature-wise centering and feature-wise standard normalisation were carried out (featurewise_center=True and feature-wise_std_normalization=True). This required taking each image's mean value from the dataset and dividing the result by the dataset's standard deviation. These procedures guarantee that the pixel values have a unit variance and a zero mean, which helps speed up model convergence.

In the case of YOLOv8, data augmentation is not explicitly required because the Ultralytics module, which is used for YOLOv8 implementation, already incorporates default

or preset data augmentation techniques. The Ultralytics package, specifically designed for object detection tasks with YOLOv8, integrates data augmentation functionality within its implementation. This indicates that when training, the package instantly and automatically applies a variety of data augmentation techniques to the input photos.

5.1.1 YOLOv8 Directory Structuring

After the dataset was prepared and labelled following the annotation process described in Chapter 4.3, the data needed to be organised to be compatible with the YOLOv8 framework. The dataset was restructured into the following folder structure in the figure 5.1.

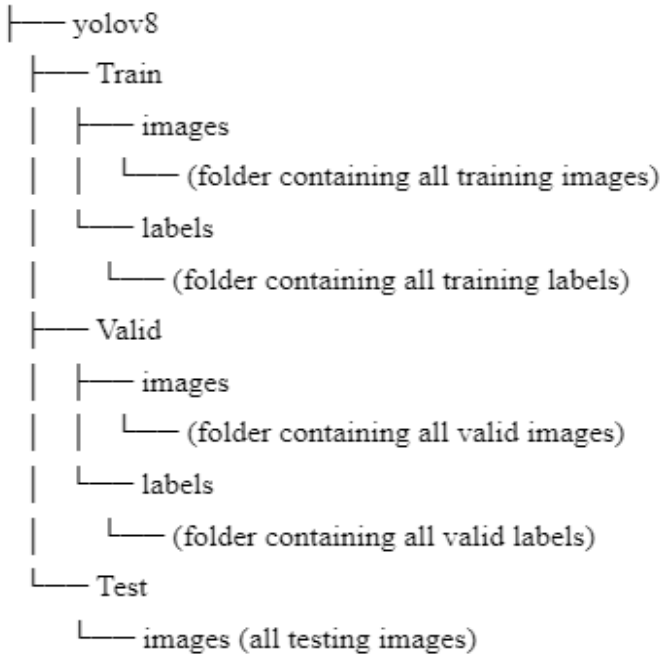


Figure 5.1 YOLOv8 folder structure

The "yolov8" directory acts as the dataset's root directory in this arrangement. The data is divided into the "Train", "Valid" and "Test" folders within the "yolov8" directory. The "Train" directory contains the training data, including the "images" folder that holds all the training images and the "labels" folder that contains the corresponding training labels.

Similarly, the "Valid" directory contains the validation data, which holds the respective images and labels for model validation. The "Test" directory contains the test data which holds all of the testing images. Once the dataset was organised in this structure, a custom configuration file called "dataset.yaml" was created to ensure that the data is correctly prepared and preprocessed to meet the input specifications of the YOLOv8 model.

5.2 Model Architecture and Training

5.2.1 InceptionV3

```
Python
#Load the InceptionV3 network

baseModel=InceptionV3(weights="imagenet",include_top=False,
input_tensor=Input(shape=(IMG_WIDTH, IMG_HEIGHT, 3)))
```

The base model used is InceptionV3, which is loaded with pre-trained weights from the "imagenet" dataset. The completely linked layers at the top of the network are omitted since the “include_top” option is set to False. The head of the model, which will be positioned on top of the base model, is then built. An AveragePooling2D layer with a pool size of (5, 5) receives the output from the base model, which decreases the feature maps' spatial dimensions. Using the Flatten layer, the output is then flattened to produce a one-dimensional vector.

The model gains non-linearity via the addition of a dense layer with 64 units and ReLU activation. A Dropout layer with a rate of 0.5 is used to avoid overfitting. Finally, a dense layer is added, with the number of units equal to the number of classes in the training dataset, and class probabilities are generated using softmax activation. By designating the inputs as the input tensor of the base model and the outputs as the headModel, the overall

model is produced. Training will be conducted using this model. A loop is used to set the trainable property of each layer in the baseModel to False, which prevents the base model's layers from changing while training. Categorical cross-entropy is used as the loss function and the Adam optimizer to compile the model. The accuracy metric is also specified for evaluation during training.

5.2.2 MobileNetV2

```
Python
#Load the MobileNetV2 network

baseModel=MobileNetV2(weights="imagenet",include_top=False,

input_tensor=Input(shape=(IMG_WIDTH, IMG_HEIGHT, 3)))
```

The same model architecture as InceptionV3 was applied to train the MobileNetV2 model with a slight modification as MobileNetV2 was used as the base model rather than InceptionV3 as shown in the code block above. The model's head and the rest of its construction were left untouched. This method made it possible to evaluate and contrast the two models in a consistent manner.

```
Python
model.compile(loss="categorical_crossentropy", optimizer='sgd',

metrics=["accuracy"])
```

To get a comprehensive evaluation, the same architecture is used and while this time compiling the model using SGD optimizer for both InceptionV3 and MobileNetV2 models. Compared to the Adam optimizer, this optimizer is different. The training process can explore a diverse optimization landscape by using SGD to build the model, possibly resulting in various model performance and the identification of the best-performing model. This

thorough analysis allows for a fair comparison of the two models and sheds light on their respective capabilities and performance on the given object detection or classification task.

5.2.3 YOLOv8

The starting weights for the network were taken from a pretrained model called "YOLOv8l" that was retrieved from the official Ultralytics github repository. This model was used to train the YOLOv8 model. YOLOv8l represents the large variant of the YOLOv8 architecture, providing a more powerful and accurate model for object detection. To initiate the training process, the Anaconda command-line interface (CLI) was utilised.

The precise CLI command that was employed is depicted in figure 5.2. With 10 epochs, the "yolov8l.pt" pretrained model as the base, an input picture size of 640x640 pixels, and a batch size of 8, this command started the training process.

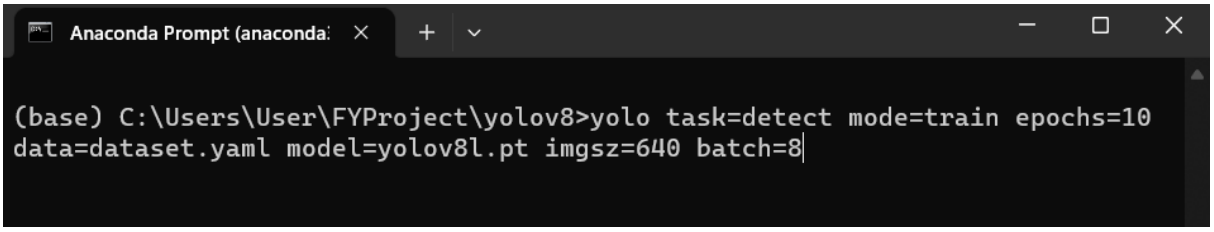


Figure 5.2 YOLOv8 train command

In order to mitigate potential CUDA out-of-memory issues, a lower batch size was set. By reducing the batch size, the GPU memory consumption during training was limited, allowing the training process to run smoothly without encountering memory constraints. Upon completion of the training, the best weight file, best.pt which can be found in ‘runs\detect\train\weights’ folder, was copied and paste into the yolov8 root folder and renamed it as “yolov8l_custom.pt” as it will be easier to use it when testing the model.

5.3 Model Testing

5.3.1 InceptionV3 and MobileNetV2

The test set was utilised for assessment in order to judge the effectiveness and generalisation skills of the trained InceptionV3 and MobileNetV2 models. The test set, which was distinct from the training and validation data, acts as an objective indicator of how well the models are able to identify or categorise unseen pictures. The trained InceptionV3 model was given the test set, and predictions were generated for each image. The accuracy, precision, and recall assessment metrics were then calculated by contrasting these predictions with the ground truth labels. The models' overall performance and their capacity to accurately identify or recognise items in the test set are revealed by the evaluation metrics.

The MobileNetV2 model was similarly evaluated using the same test data. Predictions were made after the photos were run through the trained MobileNetV2 model. To determine the same set of assessment measures, the predicted labels were compared to the ground truth labels. Both models' evaluation outcomes were noted and examined. In the analysis, the performance measures for InceptionV3 and MobileNetV2 were compared, including accuracy, precision and recall. Which model performed better on the provided test set was determined using the results. Additionally, based on the evaluation results, any insights on possible strengths, shortcomings, or opportunities for growth for each model were highlighted. The report rigorously tests the trained InceptionV3 and MobileNetV2 models on the independent test set to provide an objective evaluation, providing useful insights into their performance and applicability for the object detection or classification task. The superior model between these two is also tested on highlight video and further discussion will be covered in chapter 6.

5.3.2 YOLOv8

Two testing scenarios, testing on a highlight video and testing with the test set were carried out to gauge how well the trained YOLOv8 model performed. The specific CLI command that was employed is shown in figure 5.3 to test the YOLOv8 model with the test set. With the test dataset as the source and the following settings: prediction mode, use of the "yolov8l_custom.pt" trained model, presentation of the predicted results (show=True), confidence threshold set to 0.5 (conf=0.5), and display of the predicted results. The trained YOLOv8 model then processed the test images, detecting and localizing logos of interest according to the specified confidence threshold. The predicted outcomes were shown for visual examination and analysis.

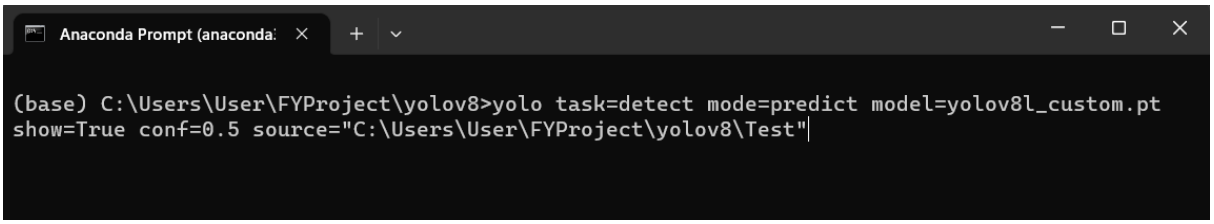


Figure 5.3 YOLOv8 predict testing on Test set command

Another CLI command as shown in figure 5.4, was used to test the YOLOv8 model on a highlight video. The source of the video was ‘testVideo.mp4’ which is the highlight badminton video. This command processed the supplied highlight video as the source and carried out the detection task in prediction mode using the "yolov8l_custom.pt" trained model. Based on the recognised patterns and characteristics, the YOLOv8 model examined the frames of the video, identifying and locating logos of interest. Each frame's predictions identified the logo and their corresponding bounding boxes.

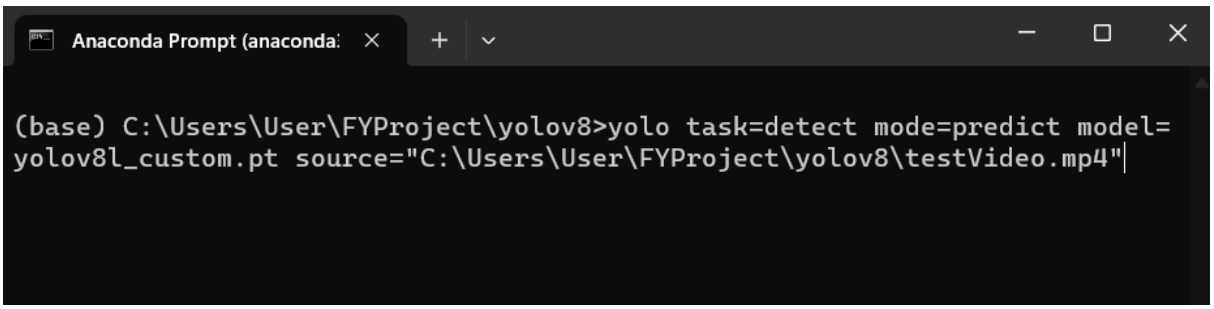


Figure 5.4 YOLOv8 predict testing on video command

The 'runs\detect' folder contained all of the model's predicted outcomes. The prediction logos, their respective confidence ratings, and the relevant bounding boxes were all included in these prediction outputs. A record of the model's performance on the test set and the highlight video was available in the saved results.

In summary, the model testing section for YOLOv8 involved testing the model on a test set and a highlight video. CLI commands were used to initiate the detection tasks, and the resulting predictions were stored for further analysis. This assessment offered information on the model's object detection performance, enabling an evaluation of its precision and efficacy in various settings.

Chapter 6 : Evaluation of Findings

6.1 Data Pre-processing

Rescaling the pixel values, using data augmentation methods including flipping, shearing, zooming, and rotating, as well as feature-wise centering and standard normalisation, were all phases in the data pre-processing process. These pre-processing steps help ensure that the input data is appropriately prepared for training the InceptionV3 and MobileNetV2 models. For the YOLOv8, by providing the proper path for the dataset folder and adjusting the number of classes and their names, the model parameters can be kept and easier to manage while training or testing the model.

6.2 Model Training and Evaluation

6.2.1 Learning Curves

The MobileNetV2, InceptionV3 and YOLOv8 models' accuracy and loss function were tracked and recorded during the training procedure. To see how these measures changed across the training epochs, learning curve graphs were created as shown in Figure 6.1, 6.2, 6.3 and 6.4. Figure 6.4 shows the YOLOv8's result graphs after the model training.

I. Adam Optimizer



Figure 6.1 Learning curve of MobileNetV2 using Adam optimizer

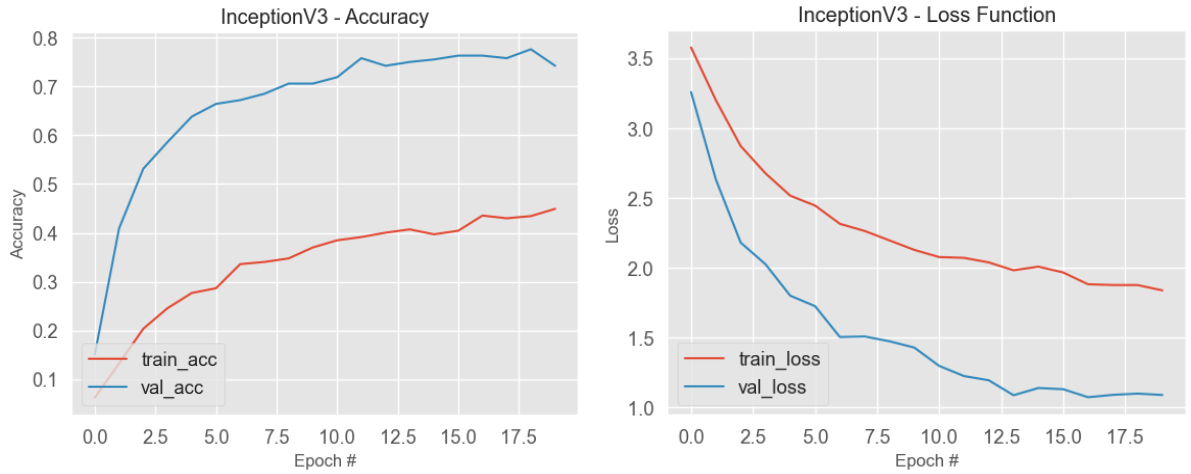


Figure 6.2 Learning curve of InceptionV3 using Adam optimizer

It was noted that both models' loss values consistently decreased as the number of training epochs rose for the loss function. InceptionV3 stood out, nevertheless, for having a considerably smaller validation loss than training loss. This shows that InceptionV3 did well on the validation set and had better generalisation than the training set, with a smaller loss. MobileNetV2, on the other hand, showed a lower difference between the training and validation loss curves, indicating that the training and validation sets performed similarly.

Regarding accuracy, both models showed an upward trend, indicating an improvement in their ability to classify or detect objects correctly. Interestingly, InceptionV3 achieved significantly higher validation accuracy compared to the training accuracy, resulting in a large gap between the curves. This shows that InceptionV3 performed better on the validation set due to its superior ability to generalise its learnt features to fresh, unknown data. MobileNetV2, on the other hand, had a much higher accuracy than InceptionV3 and showed a lower difference in the accuracy curves for training and validation, demonstrating a more consistent performance throughout both sets.

II. SGD Optimizer



Figure 6.3 Learning curve of MobileNetV2 using SGD optimizer

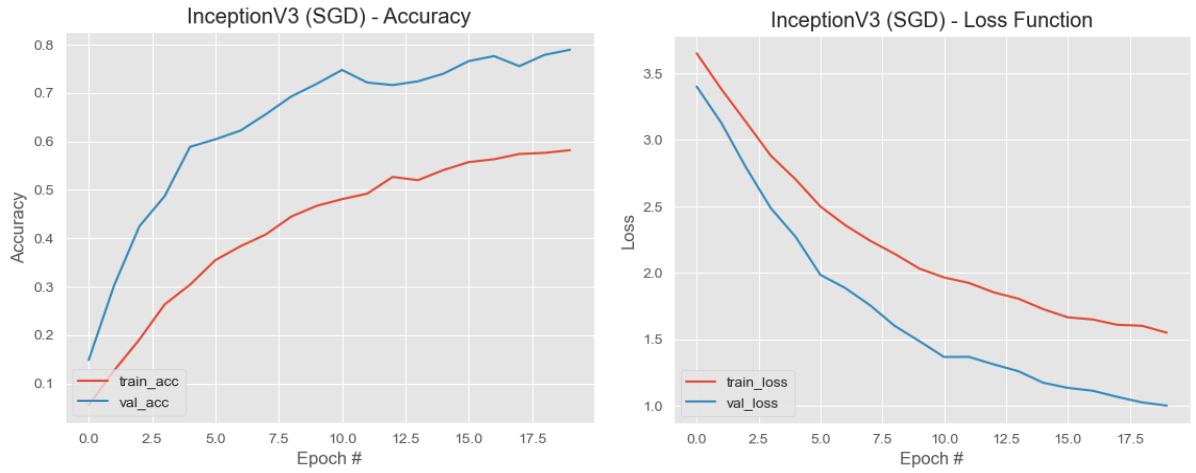


Figure 6.4 Learning curve of InceptionV3 using SGD optimizer

It was shown that both models consistently reduced their training loss values throughout training for the loss function. It is interesting to note that for both MobileNetV2 and InceptionV3, the validation loss was consistently less than the corresponding training loss. The fact that both models performed well on unseen validation data suggests that both models have improved generalisation capabilities. It is important to note that MobileNetV2 outperformed InceptionV3 in terms of validation loss, indicating that it had greater generalisation in terms of minimising the loss on unseen data.

Both models improved in accuracy, indicating an increase in their capacity to accurately identify or detect things. Notably, both MobileNetV2 and InceptionV3 routinely beat their training accuracy in terms of validation accuracy. This suggests that both models performed exceptionally well at applying their honed characteristics to fresh, unexplored data. Similar to the loss curves, MobileNetV2 outperformed InceptionV3 in terms of validation accuracy, indicating a superior generalisation capacity.

III. YOLOv8

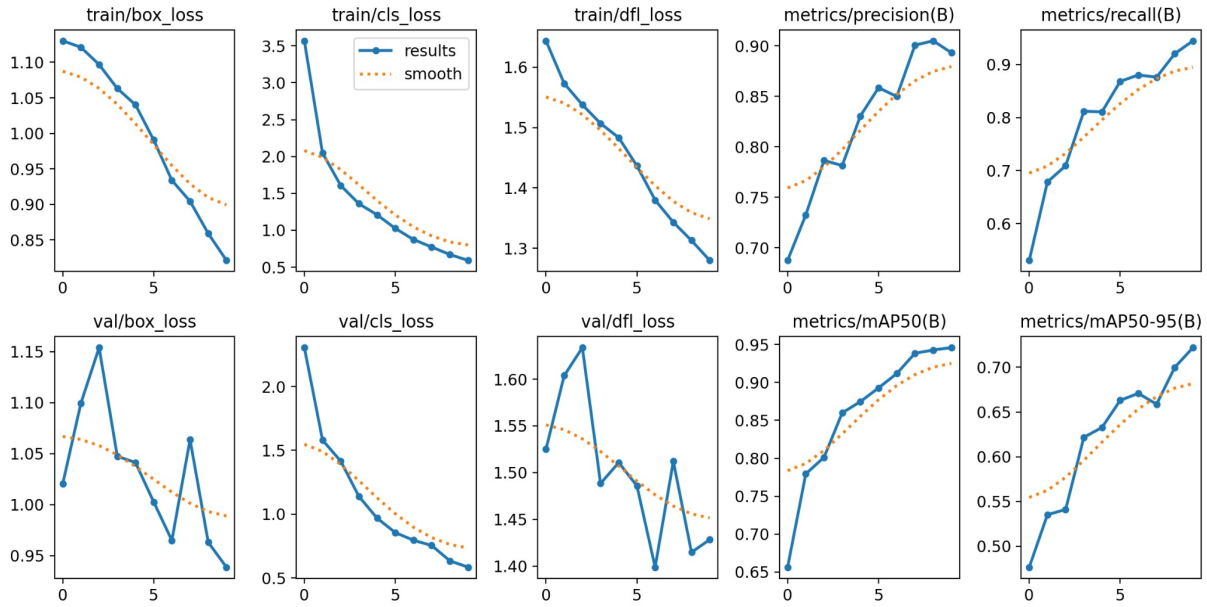


Figure 6.5 YOLOv8 plots over the training epochs for the train and valid set

There are three types of loss shown in figure 6.4 which are box loss, classification loss and dual focal loss. The box loss is the loss experienced while attempting to forecast the bounding box coordinates of an object. It calculates the difference in bounding box coordinates between the expected and actual bounding boxes. The model's ability to precisely localise logos inside the images or videos is shown by a decreased box loss. Classification loss measures the inaccuracy in assigning objects to particular classes. It calculates the discrepancy between the actual class labels and the expected class probabilities. A reduced classification loss indicates that the model can confidently categorise logos in the proper category. In YOLOv8, a specialised loss function called dual focal loss is used to deal with the problem of unbalanced data during object detection. By giving such instances more weight, it highlights how crucial it is to find uncommon or difficult objects. Improved logo detection accuracy is a result of the dual focal loss, especially for things that are less often represented in the collection.

The accuracy, recall, and mAP values showed a rising trend during training, demonstrating the model's progress in reliably recognising objects over the training epochs. Even if there may have been a few minor variations, the trajectory as a whole demonstrated steady advancement. These measurements showed that as training progressed, the model became more and more adept at handling object detection tasks.

6.2.2 Comparison of Model Performance

Table 6.1 shows the accuracy, precision, and recall of the three models, InceptionV3, MobileNetV2 trained with two different optimizers (Adam and SGD) and YOLOv8.

Table 6.1 Results of the Deep Learning Models

Model	**Optimizer**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**
InceptionV3	Adam	70.75	73	71
	SGD	73.75	77	74
MobileNetV2	Adam	81.5	83	82
	SGD	80.25	81	80
YOLOv8	Auto (Adam)	mAP: 94.6	89.4	94.4

A comparison of the accuracy, precision, and recall of the InceptionV3, MobileNetV2, and YOLOv8 models produced some intriguing results. The mAP score of YOLOv8 is 94.6%. This high mAP score reflects the model's strong performance in properly recognising and localising items, notably logos, within the dataset. YOLOv8 outperformed InceptionV3 and MobileNetV2 in accurately identifying and localising logos inside images and videos, demonstrating superior object detection capabilities. MobileNetV2 with the Adam optimizer came in second place with an accuracy score of 81.5%, while InceptionV3 with the Adam optimizer attained an accuracy score of 70.75%.

YOLOv8 maintained a high precision score of 89.4%, demonstrating its capacity to recognise and categorise items effectively while producing few false positives. With an precision score of 83%, MobileNetV2 proved its reliability for object detection systems. When utilising the Adam optimizer, InceptionV3 obtained a precision score of 73%, indicating a little inferior precision compared to the other models.

With a recall score of 94.4%, YOLOv8 demonstrated exceptional sensitivity in recognising objects and reduced false negatives. With a recall score of 82%, MobileNetV2 proved its capacity to identify a sizable proportion of true positives. Utilising the Adam optimizer, InceptionV3 achieved a recall score of 71%, which is the lowest among the other models. In terms of precision, and recall, YOLOv8 consistently surpassed both InceptionV3 and MobileNetV2. Due to its distinct object detection design and optimisation methods, it performs exceptionally well. While InceptionV3 exhibited substantially poorer performance across all parameters, MobileNetV2 also showed consistent performance.

6.3 Model Testing and Evaluation

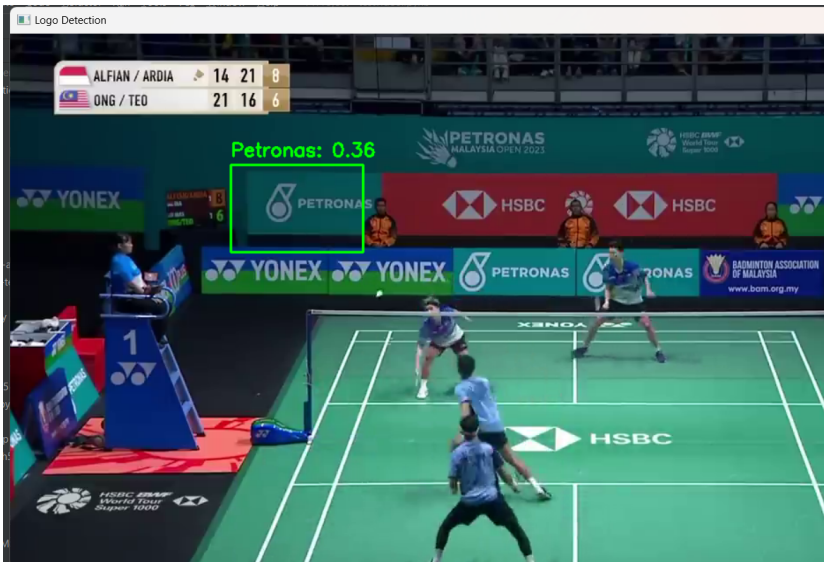


Figure 6.6 The detection results of MobileNetV2 trained with Adam model

Additionally, the MobileNetV2 trained with Adam model are chosen to be tested on highlight videos to evaluate their performance in real-world scenarios. However, it was observed that the models cannot accurately detect the logos in the videos. This discrepancy can be attributed to the fact that these models were not trained specifically with bounding box annotations, which provide precise spatial information about the location of the logos in the images.

Even though MobileNetV2 models are capable of learning and recognizing objects, including logos, and do in fact require annotations to be trained, this model does not require training data that includes bounding box level annotations for the precise localization. In the absence of such annotations, the models rely on the inherent feature representations learned during training, which may not be sufficient for accurate logo detection in videos. It is worth noting that logo detection in videos can be a more challenging task compared to static images, as logos may appear in various sizes, orientations, and positions within the video frames. As a result, MobileNetV2 and InceptionV3 are not suitable for real-time object detection in video but are really good at classification tasks.

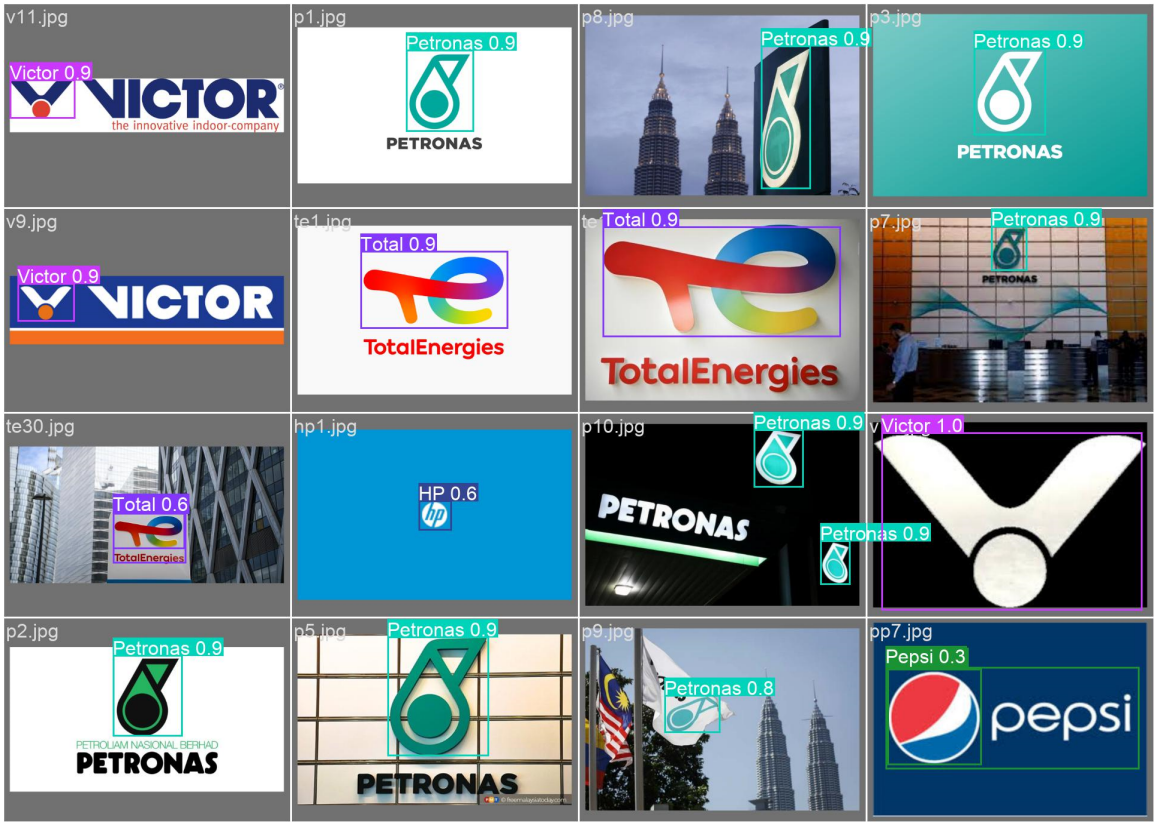


Figure 6.7 The predicted detection results of YOLOv8

When YOLOv8 was tested on the test set, it showed excellent accuracy in predicting logos. A high degree of logo detection accuracy was achieved as a consequence of YOLOv8 predictions that were consistently in line with the ground truth labels. With the use of exact bounding box coordinates, the model effectively located and localised logos inside the test images. This accurate detection allowed for reliable recognition and classification of the logos present in the images. The sophisticated architecture and specialised training methods of YOLOv8 are responsible for its excellent performance. The model successfully distinguished brands from other visual elements because of its capacity to learn intricate patterns and characteristics.

6.4 *Experimental Analysis to Brand Recognition*

It was decided to use this model to recognise logos in badminton highlight videos in light of YOLOv8's astounding accuracy in logo prediction. By using YOLOv8 in this particular sector, the model's potential use in brand identification and its capacity to promote branding awareness were investigated. Various branding aspects, like as sponsor logos, are frequently seen on player jerseys, equipment, and stadium advertising in badminton highlight videos. These logos act as visual signifiers for brands and are essential for promoting brand identification and raising consumer awareness of branding.



Figure 6.8 The performance of YOLOv8 for detecting in the highlight video

The model was trained to find and identify these trademarks in the badminton highlight video by using YOLOv8. The extraction of useful information on brand exposure, frequency, and placement can be made easier by precisely recognising and localising the logos. Insights into the efficacy of brand recognition utilising YOLOv8 might be obtained through this experimental analysis. The model's precise logo identification skills made it possible to quantify the exposure and visibility of various companies throughout the highlight video. By analysing the frequency and duration of logo appearances, valuable data that can be utilised to measure the impact of branding efforts during badminton events can be obtained.

Furthermore, YOLOv8's real-time logo detection within the highlight videos allows companies and marketers to receive fast feedback. These feedbacks may be used to evaluate the success of their branding tactics, plan intelligently for next campaigns, and raise general brand recognition among the target market. The capability of this methodology to contribute to branding awareness has been proved by adding YOLOv8 into the brand recognition pipeline for badminton highlight videos. The precise recognition and analysis of logos provide actionable insights for brands, enabling them to evaluate their brand awareness, measure audience engagement, and adjust their marketing tactics as necessary.

Chapter 7 : Conclusion

In this project, the effectiveness of three well-known models for object detection, InceptionV3, MobileNetV2, and YOLOv8 was examined and compared. In the context of logo detection and brand recognition, the evaluation and analysis of these models provided insight into their capabilities, strengths, and limits.

Our findings show that YOLOv8 outperformed both InceptionV3 and MobileNetV2 in terms of performance across multiple metrics. With mAP scores of 94.6%, 89.4% precision, and 94.4% recall, YOLOv8 demonstrated its exceptional object detection capabilities. It is an attractive option for brand recognition tasks due to its capability to reliably identify and categorise logos, minimise false positives, and capture a large percentage of true positives. Based on these findings, it is evident that YOLOv8, with its state-of-the-art architecture and exceptional performance, is the model that is best suited for logo detection and brand recognition tasks. Due to its excellent accuracy, precision, and recall scores, it has the ability to be used in the real world and help businesses efficiently identify and detect their brands across a variety of visual media.

Future research can concentrate on developing a dataset that only includes all sponsorships of BWF badminton events and a dashboard application that presents comprehensive brand detection reports. Insights into logo detection, brand exposure, and audience engagement can be provided in real-time by this application, which will be an invaluable tool for marketing teams. The marketing team may make informed decisions, improve branding tactics, and adjust marketing campaigns by visualising the data and trends.

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APPENDICES

Appendix A: Research Paper submitted to Citic 2023 and Rejected

Comparison of Deep Learning Methods for Logo Detection and Recognition from Badminton Highlight Video

Abstract. Brand exposure is particularly important for sponsors and advertisers who want to measure the effectiveness of their advertising campaigns. Logo detection and recognition is an approach to detect and recognise the logos and brands of sponsors and advertisers in badminton sports. This study will be useful to provide valuable insights into brand exposure and help to measure the effectiveness of advertising the brands in a campaign. In this paper, the focus is to study and analyze existing Convolutional Neural Network (CNN) methods for Brand Logo detection as we compared two different deep learning-based methods (i.e. MobileNetV2, InceptionV3) for logo detection. The models are being trained using images and a highlight badminton video is fed into the trained model to showcase its performance. As we also highlight the significance of evaluating the performance of the method using multiple metrics such as accuracy, Recall, Precision, and F1 score in order to have a thorough understanding of the method. As a result, this experiment shows how well CNN's approaches for brand recognition and logo detection work.

Keywords: Logo Detection and Recognition · Deep Learning · CNN · Badminton Highlight Video · Comprehensive Evaluation

1. Introduction

A customer must be exposed to a brand's favorable qualities on a frequent basis in order to develop a perception or image of the brand in their mind. This process is known as "brand recognition". Understanding how customers feel about the brand, how they place its value, what its brand image is like, how much brand knowledge they have, etc., is crucial for businesses.

Brand recognition is the main objective of advertising during the first few months a product is introduced to the market. A brand can be acknowledged even if the consumer does not always recognise its name. It actually means that after seeing a company's visual package graphics, consumers can react to them and let a company know how conscious of the brand they are. Customers are said to have successfully recognised a brand when they can do so without being given the name of the business directly by utilising visual or aural clues such as logos, slogans, packaging, colors, or jingles. Companies regularly do market research to evaluate the effectiveness of their brand awareness campaigns. The ultimate marketing goal is to make the target market think about the product when they are ready to buy in order to distinguish the brand from the competitors. High brand recognition is connected with strong brand equity.

Logo detection in images and videos is an important task for a variety of applications, including traffic-monitoring systems, vehicle logo detection, copyright infringement detection, and brand recognition. Further analysis can be performed, for example, to monitor a brand’s presence among consumers. Increasing brand recognition is essential to the business's success. In other words, the more

successful a brand or company is, the more customer brand awareness or recognition there is. This research will look at how effective brand identification and logo detection are accomplished using deep learning techniques like convolutional neural networks (CNN) as we performed a comparative study between the existing deep learning-based methods.

2. Literature review

2.1. Wordmark Logo vs Graphic Logo



Fig. 1: Wordmark Logos vs Graphic Logos. Taken from Brackett (2019)

A Graphic Logo is a form of logo that combines text with visuals or symbols. The key issue is their abstract and frequently complex design, which can make it challenging for a model to correctly detect and classify them. Additionally, graphic type logos may be used in a variety of contexts, such as on products or in advertisements, which can result in variations in color, size, and orientation that make the detection process more difficult. Wordmark Logos are a particular kind of logo that simply has text and no additional symbols or graphics. The biggest difficulty is the text's inconsistent look, which includes font, size, and color. Because of this, word type logos may be challenging for a model to reliably identify and recognise, especially if they are displayed in unusual orientations or alongside other text or pictures. Wordmark logos may also contain stylized text or other design features, which makes the detecting procedure even more difficult.

2.2. Overview on Deep Learning method CNN

Convolutional neural networks are a subtype of neural networks that are particularly skilled at processing input having a grid-like design, such as pictures. CNNs and ConvNets are other names for convolutional neural networks. For deep learning algorithms, a CNN is a specific kind of network design that is used for tasks like image recognition and pixel data processing. Despite the fact that deep learning uses a variety of neural network types, CNNs remain the preferred network design for recognising and classifying objects. There are several reasons why CNNs are the preferred network architecture for identifying and recognizing objects. One reason is that CNNs are able to automatically learn a hierarchy of features from raw data, such as pixels in an image. This is done through a process called convolution, where a small, learnable filter is applied to the input data to extract features at different scales and locations. These features are then passed through a series of non-linear transformations, such as activation functions, to

extract more complex features. This process is repeated multiple times, forming a hierarchy of features that can be used to classify or recognize objects in the input data.

Another reason why CNNs are effective for object detection and recognition is that they are able to handle large amounts of data and can be trained on large datasets. This is important because object recognition tasks often require a large number of examples in order to learn the variations and variations in object appearance. CNNs are also able to learn from data that has a high level of variability, such as images that are taken from different angles or under different lighting conditions. This makes them well-suited for tasks such as object detection in real-world images.

2.3. Recent Works

In recent works of logo detection and recognition, most authors proposed a model or method that is CNN-based. Y. Meng and S. Hou et al (2021) created a strong baseline technique dubbed "obtaining sufficient features for logo" (OSF-Logo) to address the issues with multiscale objects, large-scale categories, viewpoint logos, and part deformation. This method is mainly divided into four parts as shown in the figure below.

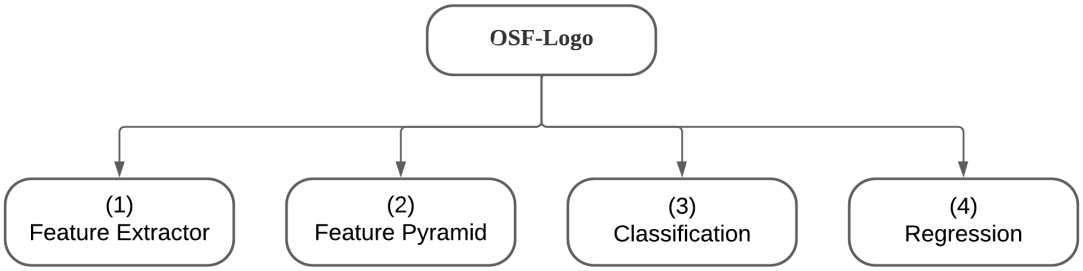


Fig. 2: Four Parts of OSF-Logo method. Taken from Y. Meng et. al. (2021)

The first part is Feature Extractor specifically using the Regulated Deformable Convolution (RDC) module with offsets and amplitudes to more convolution layers. Second part is Feature Pyramid, where the authors constructed Content-Aware ReAssembly Features in FPN (FPN-CARAFE) (CFPN) with an up-sampling operator for combining data into a sizable receptive field to detect multi-scale logos. Third part is Classification. With the help of dynamic content perception, the suggested CFPN method achieves feature reorganization in order to extract the features of small targets and regress large-scale logos. The main idea behind CARAFE is to improve the efficiency and effectiveness of convolutional neural networks (CNNs) for image processing tasks by using a dynamic, content-aware approach to reassembling feature maps. In traditional CNNs, feature maps are typically generated by applying a series of convolutional filters to the input data. These feature maps are then passed through a series of non-linear transformations, such as activation functions, to extract more complex features.

The Kernel Prediction module is used to create the recombined computation's kernel while the Feature Recombination module is to determine the points that correspond to each pixel on the original feature map. Last part is Regression, Y. Meng adds DIoU loss to regression as regression loss for precise logo detection. Similarly, Hang Su and S. Gong et al (2021) proposed a Multi-Perspective Cross-Class (MPCC) which aligns the feature distribution among synthetic logo images, real logo images, and non-logo object photos by investigating unsupervised domain adaptation. Extensive tests on the publicly available QMUL-OpenLogo benchmark demonstrate the performance advantage of the proposed MPCC

method for open logo detection over cutting-edge alternative approaches. The main concept is to apply the understanding of fully supervised logo classes to 1-shot icon supervised logo classes.

Ahmed Alsheikhy et al (2020) proposed an architecture for logo recognition. The authors are interested in effective representations that can be accomplished with a limited set of parameters as they tackle recognition with a small amount of original data. So the authors explored the Densely Connected Convolutional Networks (DenseNet) to get efficient results. A DenseNet is a kind of CNN that uses Dense Blocks to directly connect all layers (with matching feature-map sizes) with one another in order to create dense connections between them.

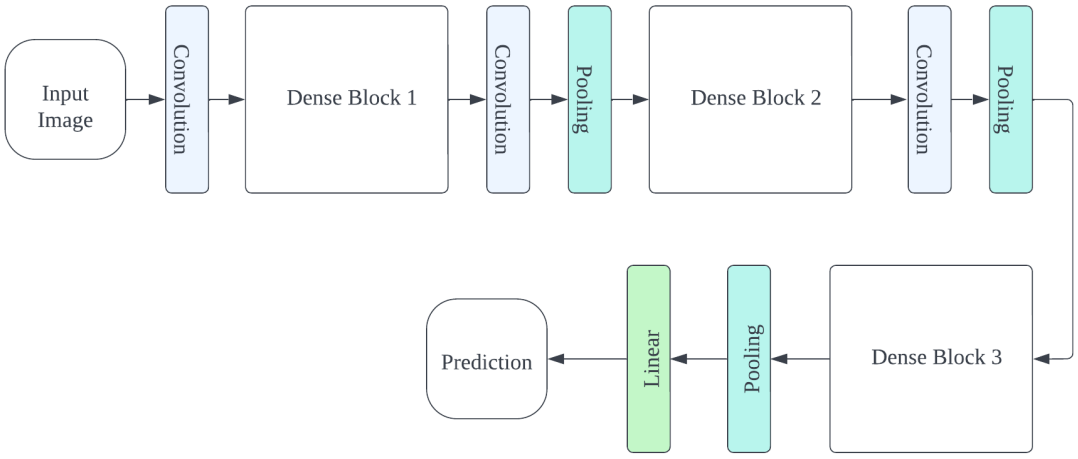


Fig. 3: DenseNet architecture. Taken from Ahmed et. al. (2020)

A collection of convolutional layers with extensive interconnections is referred to as a dense block. All outputs from earlier layers are given to the current layer as input. Convolution layers are packed into a single dense block, followed by ReLU activations layers and Batch Normalisation Layers which is to reduce the size of feature maps. Instead of using the classes from the original ImageNet dataset, Ahmed et al. used the transfer learning technique to configure the output layer for the logos classes. The reason is to enable the reuse of Deep CNN models that have already been created without the need for extensive computing. Plus it also helps prevent overfitting when accelerating network training. The weights of the custom-made network were initialized with a pre-trained DenseNet model on the ImageNet Dataset. Then, the network was fine-tuned on the FlickerLogo32 training set.

K. Kumar J. P. and Supriya M. C. (2021) also proposed a CNN-based system for logo detection in television programs.The suggested system comes with a pipeline for logo detection and classification in TV broadcast and broadband content that demonstrates how to use the object detection algorithm Single Shot Detector (SSD) on a TV broadcast channel logo dataset that has undergone significant changes to better represent various logo conditions. Generally speaking, the suggested system consists of two pieces. The first section focuses on the extraction of frames, and to create fake data, different data augmentation techniques are used with the retrieved broadcast logo. This data augmentation is done to increase the trained network's coverage of various logo-types in terms of size, form, and visibility. In the second part, Single shot detector (SSD) architecture is used for single forward pass detection and classification. SSD employs the VGG16 design without taking into account the fully connected layers. The VGG16 architecture was selected as the base due to its consistently good performance for classification tasks. The

number 16 in the name VGG indicates the fact that it is a 16 layers deep neural network (VGGnet). This indicates that the VGG16 network is quite large, with a total of over 138 million parameters. But instead of using VGG16 fully connected layers, a group of convolutional layers has been added, enabling input size reduction at each layer as well as feature extraction at various scales. The frame that was intercepted is sent to the VGG16 network model, which produces the feature extraction. These features are then sent to several convolutional layers and convolutional filters, which are used to identify and categorize the item.

Amy J.C. Trappey and Charles V. Trappey et al (2022), integrates two deep learning models to achieve the goal of assuring copyrighted Trademark (TM) logo protection. The first model is Logo Detection and Localization (LDL), which is used to clip trademark-like images from intricate online product photography, which may include shots of various product configurations or even wearable clothing exhibited on a human model. For comparison to a database of registered trademarks, only the cropped image of the logo is required. The second model, utilizing the cropped logos for the TM similarity measurement (TMSM) model, compares the published TM pictures. The universal logo locator for intelligent logo image cropping is Yolo v4. For intelligent trademark similarity analysis, the triplet convolutional neural network ResNet50 model is used. The models are trained using an image dataset that combines LogoDet-3k picture samples with photos retrieved from the internet.

3. Methodology

Our methodology involves training a deep learning model using images to recognize specific patterns and objects. The dataset consists of thousands of logo images that are carefully curated and annotated to provide the necessary information for the model to learn from. The existing convolutional neural network (CNN) models that are being used are well-suited for image recognition tasks due to its ability to extract features and patterns from images. Once the model is trained, the model will be implemented on a highlight badminton video to showcase its performance. The video will contain several scenes that feature the logos the model was trained on, and the model will be able to identify and highlight those logos in real-time. This will demonstrate the effectiveness and accuracy of the model in a practical and visually engaging way.

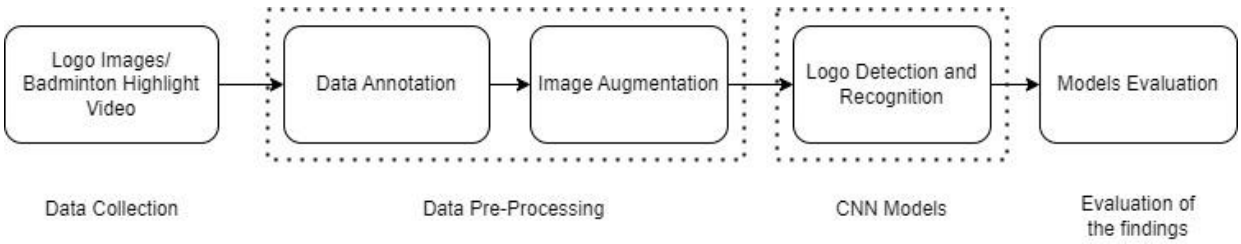


Fig. 4: General Methodology Framework

3.1. Data Collection

For the creation of deep learning models, especially those requiring the detection and recognition of objects, the availability of datasets is essential. However, there aren't many datasets available in Malaysia that can be used to recognise the logos of well-known companies in Malaysia, particularly those that support badminton competitions. A new image dataset is suggested to be produced in order to get around this restriction. This custom dataset consists of all 40 logos from 40 different brands across the four categories of electronics, food, medicine, and transportation that all of the logos are well-known brands in

Malaysia. The collection, which contains more than 5,000 photos, are compiled from several web sources. The compilation of this dataset makes it possible to create models for logo identification and recognition that are more precise and trustworthy for well-known brands in Malaysia.

3.2. Data Pre-processing

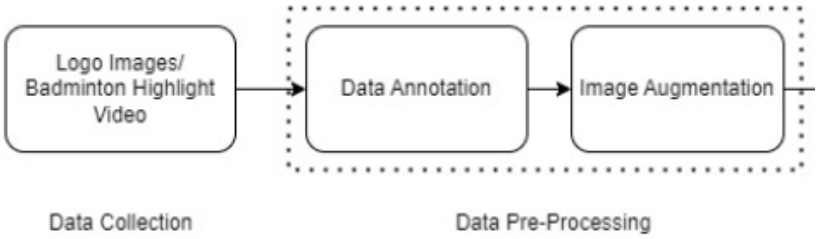


Fig. 5: First phase of Methodology Framework

The raw photos are processed in the data preprocessing phase of the logo identification and recognition project so that they may be input into the machine learning model. First, the 5,000+ images that have been gathered from different online sources are resized to a fixed size of 320 x 320 pixels. The photos are then normalized so that all of their pixel values fall between 0 and 1. This is crucial for the neural network to learn the features of the photos. Furthermore, to expand the dataset and strengthen the model's robustness, data augmentation techniques like rotation, flipping, and zooming are used.

Additionally, the images are labeled and transformed into a format that the deep learning model can learn from, such as XML, YOLO or JSON format especially for the YOLO model. Finally, to assess the model's performance on unobserved data, the preprocessed dataset is divided into 80% of training, 10% of validation, and 10% of testing sets while maintaining a balanced distribution of logos among the various classes.

3.3. Deep Learning Models

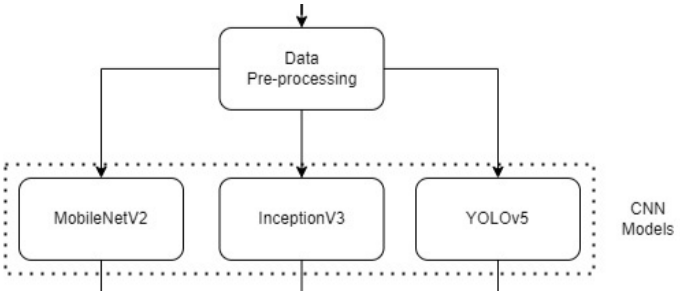


Fig. 6: Second Phase of Methodology Framework

Once the data pre-processing process is completed, the pre-processed images will be input into the different deep learning models which are MobileNetV2, InceptionV3 and maybe with YOLOv5.

3.3.1. InceptionV3

Convolutional neural network model InceptionV3 was created by Google in 2015. It is a design for a deep neural network that is mostly employed for image recognition and classification applications. The 48-layer InceptionV3 model is an adaptation of the original Inception architecture, InceptionV1 which

was introduced as GoogLeNet in 2014. On numerous benchmark datasets, including ImageNet, it is renowned for achieving state-of-the-art performance.

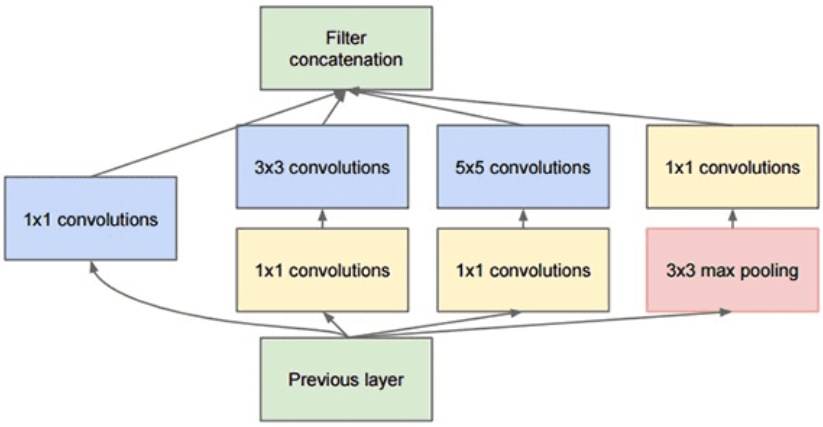


Fig. 7: The original Inception module used in GoogLeNet. Taken from Szegedy (2014)

Convolutional layers, pooling layers, fully linked layers, and auxiliary classifiers make up the InceptionV3 architecture. There are a total of 23 inception modules in the network, each of which is intended to collect characteristics at a different size. Convolutional layers of various sizes, max pooling, and average pooling processes make up the inception modules. Following concatenation, the results of these operations are fed into the subsequent layer. To reduce overfitting, the model additionally uses batch normalization and dropout methods.

InceptionV3 is renowned for its computational efficiency in addition to its accuracy. Large dataset training is made simpler by the model's speedier and more memory-efficient architecture compared to its forebears. All things considered, InceptionV3 is a strong model that has been used in a variety of computer vision applications and has shown remarkable results on a number of benchmark datasets.

3.3.2. MobileNetV2

MobileNetV2 is a popular convolutional neural network architecture that was introduced in 2018. It was created especially for embedded and mobile vision applications with constrained computing and power resources. The usage of depth-wise separable convolutions, which are computationally effective and minimize the amount of parameters in the network, is a major component of MobileNetV2. The architecture comprises a series of bottleneck layers that expand and compress the feature maps after each set of depth-wise separable convolution blocks. To enhance the gradient flow during training, the model additionally has residual connections and shortcut connections.

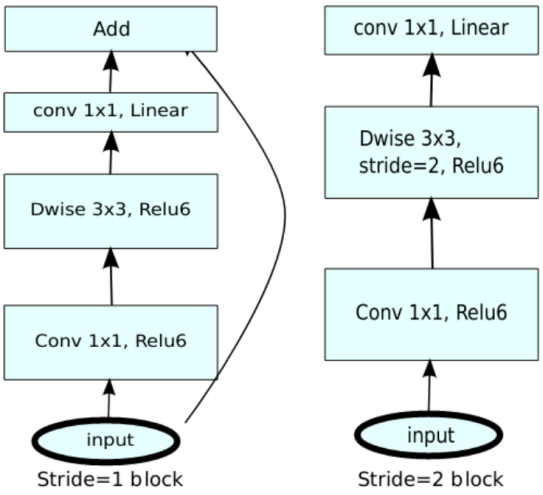


Fig. 9: MobileNetV2. Taken from Sandler (2018)

A global average pooling layer, a fully connected layer, and a softmax activation function to output the classification probabilities make up MobileNetV2's final layers. Overall, MobileNetV2 offers an excellent balance between efficiency and accuracy, which makes it a popular option for embedded and mobile vision applications.

3.4. Implementation

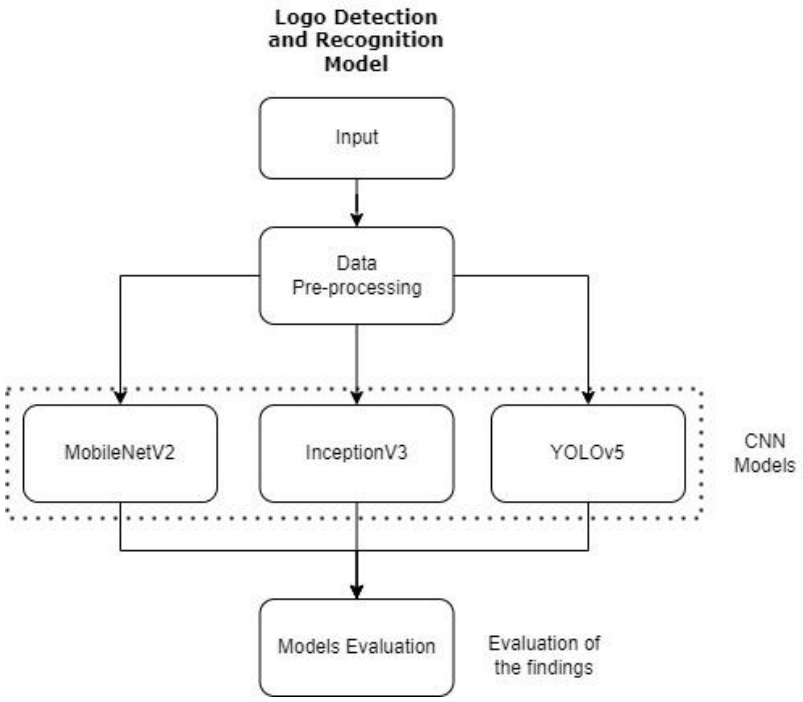


Fig. 11: Detail of Methodology Framework

The first step in the proposed workflow is to train the models using the MobileNetV2 and InceptionV3 architectures. The models will be trained on the training dataset with a batch size of 32, an Adam

optimizer, and a learning rate of 0.001. A learning rate of 0.001 is a good starting point for neural networks because it allows for gradual convergence towards the ideal solution while still enabling a reasonable learning rate. For multi-class classification applications like logo detection and recognition, categorical cross-entropy is a commonly used loss function. As the objective of this research is to categorize an image into one of many logo categories, categorical cross-entropy is an appropriate loss function. This loss function punishes incorrect predictions and encourages the model to improve by accurately assigning the right class probability by comparing the predicted class probabilities with the actual class labels. Therefore, using categorical cross-entropy as the loss function can help increase the model's accuracy in classifying logos in images.

The next step is to evaluate the trained models using the validation set. The trained models are evaluated by measuring the accuracy, precision, recall, F1 score, and mean Average Precision (mAP) for the logo detection and recognition tasks. After evaluation, the models' hyperparameters are fine-tuned by changing the learning rate to 0.0001 and apply L2 regularization with a weight decay of 0.001 to optimize the model's performance. Then, using the test set to validate the final optimized models and compare the results between the models.

To get a comprehensive evaluation, the preceding processes are repeated and while this time compiling the model using SGD optimizer. By doing so, the best-performing model for logo detection and recognition in the Malaysian context can be identified.

3.5. Evaluation Metrics

Several metrics will be employed to assess the effectiveness of the logo detection and recognition models. Firstly, the accuracy of the models will be measured, which is the ratio of correctly predicted instances to the total number of instances. This metric is important to ensure that the model is correctly identifying logos in the images. Secondly, precision will be used as a metric to measure the proportion of true positive predictions out of all positive predictions made by the model. Precision is crucial in this task to ensure that the model is not making too many false positive predictions.

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{Relevant\ Images}{Total\ Images\ Retrieved}$$

Third, recall will be utilized to calculate the percentage of accurate positive predictions among all instances of positive data that actually occurred. This measure is crucial to make sure that the model has all logos that should be included in the photos.

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{Relevant\ Images}{Total\ Relevant\ Images\ in\ Data}$$

Fourth, to provide a comprehensive assessment of the model's performance, the F1 score will be employed as a mix of precision and recall. Given that it accounts for both precision and recall, this statistic is helpful in comparing the effectiveness of various models.

$$\text{F1 score} = 2 * \frac{Precision * Recall}{Precision + Recall}$$

The performance of the logo detection and recognition models can be comprehensively assessed and compared in terms of their efficiency in identifying and localizing logos in photos.

4. Results and Discussion

Table 1 shows the accuracy, precision, recall, and F1 score of two image recognition models, InceptionV3 and MobileNetV2, trained with two different optimizers, Adam and SGD.

Table 1: Results of the Deep Learning Model

Model	Optimizer	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
InceptionV3	Adam	70.75	73	71	71
	SGD	73.75	77	74	74
MobileNetV2	Adam	81.5	83	82	81
	SGD	80.25	81	80	80

In addition to the results shown in the table, there are a few other observations that can be made. First, MobileNetV2 is a smaller model than InceptionV3. This may explain why it has higher accuracy, as smaller models are often more efficient at learning and generalizing to new data. Second, Adam is a more recent optimizer than SGD. This may also explain why MobileNetV2 trained with Adam has a higher accuracy, as Adam is generally considered to be more effective than SGD at training deep neural networks. Finally, the accuracy of both models is relatively high, which suggests that both models are capable of performing well on image recognition tasks. Overall, the results of this experiment suggest that MobileNetV2 is a promising model for image recognition tasks. It is smaller and more efficient than InceptionV3, and it can achieve comparable or even better accuracy. Additionally, MobileNetV2 is trained with Adam, which is a more recent and effective optimizer. As a result, MobileNetV2 is a good choice for image recognition tasks where speed and efficiency are important.

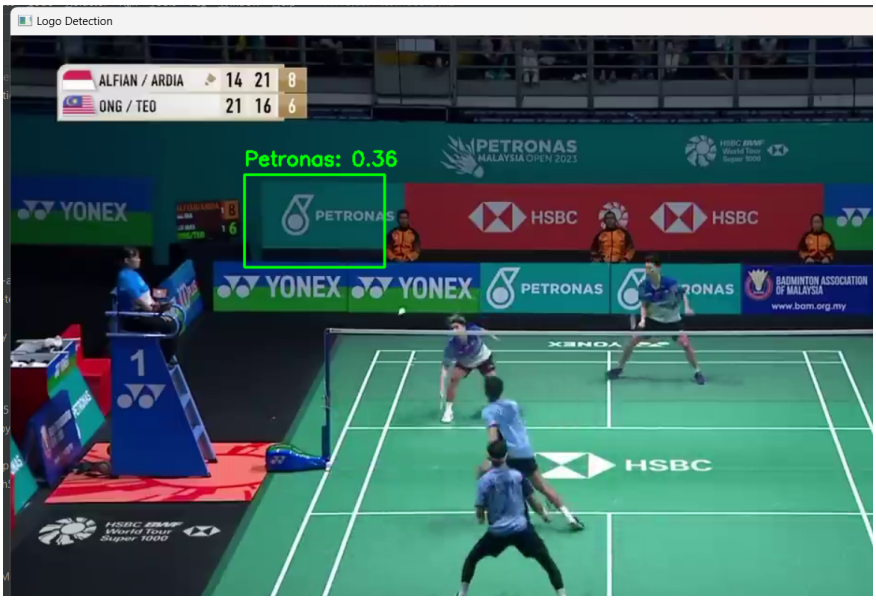


Fig. 12: The detection results using the MobileNetV2 trained with Adam model

MobileNetV2 and InceptionV3 are both deep convolutional neural networks (CNNs) that are commonly used for image classification tasks. However, they are not well-suited for object detection in video. When evaluating the model using video, as seen in Fig. 12, the detection score is not good and there are several erroneous detections. This is because these two CNNs are designed to learn features from static images, and they are not able to effectively track objects that are moving in a video. Additionally, they are computationally expensive, and they can be slow to process video frames. Plus there are a number of factors that can make it difficult for models to perform well. These factors include occlusion, deformation, and motion blur. When one object entirely or partially blocks another, it is said to be occluded. Deformation occurs when an object changes shape or size. Motion blur occurs when an object is moving quickly and the camera is not able to capture a clear image. MobileNetV2 and InceptionV3 are not designed to handle these challenges, and this can lead to errors in object detection. As a result, MobileNetV2 and InceptionV3 are not suitable for real-time object detection in video.

Image annotation is the process of manually labeling images with information about the objects they contain. This information can be used to train object detection models that are able to identify and track objects in video. Even though MobileNetV2 and InceptionV3 can be used for object detection, these models do not require image annotation to train. Instead, they are pre-trained on large datasets of images that have already been labeled with object information, which allows the models to learn to recognize these objects without the need for additional training data. In contrast, YOLO and Faster R-CNN are two other popular object detection models. These models require image annotation to train. This is because they use a technique called region proposal, which involves identifying potential regions in an image that could contain objects. The annotations are used to train the model to identify the objects in these regions. Image annotation is a time-consuming process, but it is essential for training accurate object detection models. Without image annotation, these models would not be able to learn to identify objects in new images.

Utilizing image annotation for object detection in videos offers several benefits. One of the key advantages is the improvement in accuracy that annotation brings to the object detection models. By providing explicit annotations on the objects within the images, the model gains a clearer understanding of the objects' features and can make more precise predictions. This leads to enhanced accuracy in detecting objects in video frames. Furthermore, image annotation contributes to increased flexibility in object detection. Through the annotated data, the model can learn to recognize and distinguish a wide range of objects, allowing it to adapt to different scenarios and object categories. This flexibility is crucial when dealing with diverse video content, where objects may vary in appearance, size, and context. Another advantage is the reduction in training time facilitated by image annotation. By providing initial guidance on the objects to be detected, the annotations serve as a starting point for the model's learning process. This significantly accelerates the training process, as the model has a clearer understanding of the task at hand and can quickly converge to an optimal solution.

In summary, image annotation plays a pivotal role in training accurate and flexible object detection models for videos. By improving accuracy, increasing flexibility, and reducing training time, it empowers the models to effectively detect and classify objects within video frames, opening up possibilities for various applications in computer vision and video analysis.

5. Conclusion

In this project, we compared the performance of two deep learning methods for logo detection and recognition from badminton highlight videos. The two methods were MobileNetV2 and InceptionV3. We found that MobileNetV2 was slightly better at logo detection, while InceptionV3 was slightly better at logo recognition.

The findings show the potential efficacy of deep learning techniques for locating and identifying logos in badminton highlight films. This can have important ramifications for a variety of branding objectives, such as gauging the success of marketing campaigns, locating prospective sponsorship possibilities, and monitoring brand popularity. However, the availability of a sizable and varied dataset of labeled images becomes necessary in order to further improve the performance of deep learning approaches in logo identification and recognition. Deep learning models heavily rely on the training data provided to them. If the dataset is not sufficiently extensive or diverse, the model may not be able to adequately learn and generalize to recognize logos in various contexts. Therefore, ensuring a comprehensive and diverse dataset is essential for optimizing the performance of deep learning models in logo detection and recognition tasks.

One way to improve the performance of deep learning methods for logo detection and recognition is to use image annotation. Image annotation is the process of manually labeling images with the names of the objects that they contain. This can be a time-consuming process, but it can be very effective in improving the performance of deep learning models. In the future, we plan to explore the use of image annotation to improve the performance of deep learning methods for logo detection and recognition from badminton highlight videos. We also plan to explore the use of YOLO methods like YOLOv5 or maybe the latest YOLO version, YOLOv8 for the branding purposes, such as measuring the effectiveness of marketing campaigns and identifying potential sponsorship opportunities.

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Appendix B: FYP2 Turnitin Report

ORIGINALITY REPORT

14%	9%	8%	5%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Multimedia University Student Paper	2%
2	Ye Meng, Sujuan Hou, Jing Wang, Weikuan Jia, Yuanjie Zheng, Awudu Karim. "An adaptive representation algorithm for Multi-scale logo detection", Displays, 2021 Publication	1%
3	Amy J.C. Trappey, Charles V. Trappey, Eason Lin. "Intelligent trademark recognition and similarity analysis using a two-stage transfer learning approach", Advanced Engineering Informatics, 2022 Publication	1%
4	ijiemr.org Internet Source	1%
5	Kiran Kumar Jakkur Patalappa, Supriya Maganahalli Chandramouli. "Robust recognition of logo under makeover in the context of content piracy", Global Transitions Proceedings, 2021 Publication	1%

Appendix C: FYP2 Meeting Logs

MULTIMEDIA UNIVERSITY

FACULTY OF COMPUTING & INFORMATICS

TPT3101 Final Year Project (FYP2) Meeting Log
Trimester 2, 2022/23 (Trimester ID:2220)

Meeting Date: 14 / 4 / 2023	**Meeting No.:** 1
Meeting Mode: Online via google meet	
Project ID: 2235	**Project Type:** Research-based
Project Title : Logo Detection and Recognition	
Student ID : 1191100792	**Student Name:** Raja Muhammad Darwisy Bin Raja Ahmad Al'Konee
Student Programme and Specialisation: Bachelor Degree of Computer Science (Data Science)	
Supervisor Name: Noramiza Binti Hashim	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: Revising from fyp1

(Please strike out the tasks, which are not applicable)

Details (in point form):

- New dataset collected

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Describe the description of the project's primary issue in detail
- Describe what are the FYP contribution
- Prep the dataset to start implement with at least a proposed model

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- By mistakenly reformatting the device, the whole initial custom dataset was lost.
- Discovered a comparable dataset from Kaggle to use as the solution

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



……***n.h.***……
Supervisor's Signature

……...
Student's Signature

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Co-Supervisor's Signature
(if applicable)

…….
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student must upload the soft copies of the meeting logs in Google Classroom and attach them along with final (FYP2) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee.

   This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

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TPT3101 Final Year Project (FYP2) Meeting Log
Trimester 2, 2022/23 (Trimester ID:2220)

Meeting Date: 19 / 4 / 2023	**Meeting No.:** 2
Meeting Mode: Online via google meet	
Project ID: 2235	**Project Type:** Research-based
Project Title : Logo Detection and Recognition	
Student ID : 1191100792	**Student Name:** Raja Muhammad Darwisy Bin Raja Ahmad Al'Konee
Student Programme and Specialisation: Bachelor Degree of Computer Science (Data Science)	
Supervisor Name: Noramiza Binti Hashim	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: Revising from fyp1 / Implementation

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Done describe the description of the project's primary issue in detail
- Done describe what are the FYP contribution
- Done Prep the dataset

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Implement the first 2 proposed model
- To select the final model to be employed
- To draft a conference paper

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

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4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



……***n.h.***……
Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

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2. Student must upload the soft copies of the meeting logs in Google Classroom and attach them along with final (FYP2) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee.

   This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

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TPT3101 Final Year Project (FYP2) Meeting Log
Trimester 2, 2022/23 (Trimester ID:2220)

Meeting Date: 24 / 5 / 2023	**Meeting No.:** 3
Meeting Mode: Online via google meet	
Project ID: 2235	**Project Type:** Research-based
Project Title : Logo Detection and Recognition	
Student ID : 1191100792	**Student Name:** Raja Muhammad Darwisy Bin Raja Ahmad Al'Konee
Student Programme and Specialisation: Bachelor Degree of Computer Science (Data Science)	
Supervisor Name: Noramiza Hashim	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: Implementation

(Please strike out the tasks, which are not applicable)

Details (in point form):

- The first 2 proposed model has been implemented
- The final model that to be employed has been selected
- Done draft a conference paper

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Evaluation of Findings and Research Contribution (Research-based projects)

(Please strike out the tasks, which are not applicable)

Details (in point form):

- To implement the trained models with a video
- To finalize the conference paper

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- No problem encountered

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



……***n.h.***……
Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

……
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student must upload the soft copies of the meeting logs in Google Classroom and attach them along with final (FYP2) report.
   Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee.

   This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

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TPT3101 Final Year Project (FYP2) Meeting Log
Trimester 2, 2022/23 (Trimester ID:2220)

Meeting Date: 2 / 6 / 2023	**Meeting No.:** 4
Meeting Mode: Online via google meet	
Project ID: 2235	**Project Type:** Research-based
Project Title : Logo Detection and Recognition	
Student ID : 1191100792	**Student Name:** Raja Muhammad Darwisy Bin Raja Ahmad Al'Konee
Student Programme and Specialisation: Bachelor Degree of Computer Science (Data Science)	
Supervisor Name: Noramiza Hashim	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: Implementation

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Done implement the trained models with a video
- Done finalize the conference paper

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Evaluation of Findings and Research Contribution (Research-based projects)

(Please strike out the tasks, which are not applicable)

Details (in point form):

- To submit the citic 2023 paper
- To annotate the custom dataset
- To start implement the selected final model

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- No problem encountered

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



……***n.h.***……
Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

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3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee.

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TPT3101 Final Year Project (FYP2) Meeting Log
Trimester 2, 2022/23 (Trimester ID:2220)

Meeting Date: 5 / 6 / 2023	**Meeting No.:** 5
Meeting Mode: Online via google meet	
Project ID: 2235	**Project Type:** Research-based
Project Title : Logo Detection and Recognition	
Student ID : 1191100792	**Student Name:** Raja Muhammad Darwisy Bin Raja Ahmad Al'Konee
Student Programme and Specialisation: Bachelor Degree of Computer Science (Data Science)	
Supervisor Name: Noramiza Hashim	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Implementation / Evaluation of Findings and Research Contribution (Research-based projects)
Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Done finalize the conference paper
- Done submit the citic 2023 paper

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Evaluation of Findings and Research Contribution (Research-based projects) / Research Paper (Research-based Projects)

(Please strike out the tasks, which are not applicable)

Details (in point form):

- To annotate the custom dataset
- To start implement the selected final model
- To start work on report after all of the results of the findings has been done

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- No problem encountered

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

- The low results class in the dataset need to restructure so that the model can learn properly and the results can get better when implement into test set or highlight video.



……***n.h.***……
Supervisor's Signature

……..
Student's Signature

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Co-Supervisor's Signature
(if applicable)

…….
Company Supervisor's Signature
(if applicable)

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TPT3101 Final Year Project (FYP2) Meeting Log
Trimester 2, 2022/23 (Trimester ID:2220)

Meeting Date: 6 / 7 / 2023	**Meeting No.:** 6
Meeting Mode: Online via google meet	
Project ID: 2235	**Project Type:** Research-based
Project Title : Logo Detection and Recognition	
Student ID : 1191100792	**Student Name:** Raja Muhammad Darwisy Bin Raja Ahmad Al'Konee
Student Programme and Specialisation: Bachelor Degree of Computer Science (Data Science)	
Supervisor Name: Noramiza Hashim	**Co-Supervisor Name:** **(if applicable)**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE
[Please write the details of the work done, after the last meeting]

Tasks: Implementation / Evaluation of Findings and Research Contribution / Research Paper (Research-based Projects)

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Done annotate the custom dataset
- Done implement the selected final model
- Started reporting all of the results of the findings that has been done

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]

Tasks: Draft Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- To draft the final report before email to supervisor to get comments and performing corrections based on the comments

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- No problem encountered

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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……***n.h.***……
Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

…….
Company Supervisor's Signature
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